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(54) SUBSTRATE PROCESSING SYSTEM AND TRANSFER METHOD

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(57) **ABSTRACT**

A substrate processing system includes: a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and a controller. The controller performs: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

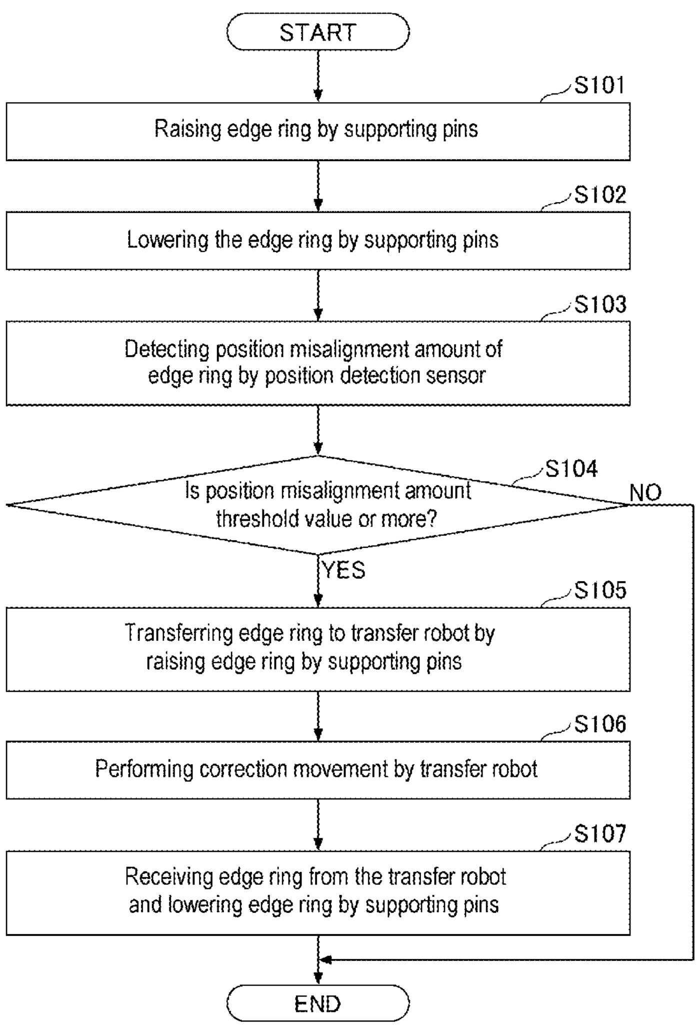


FIG. 1

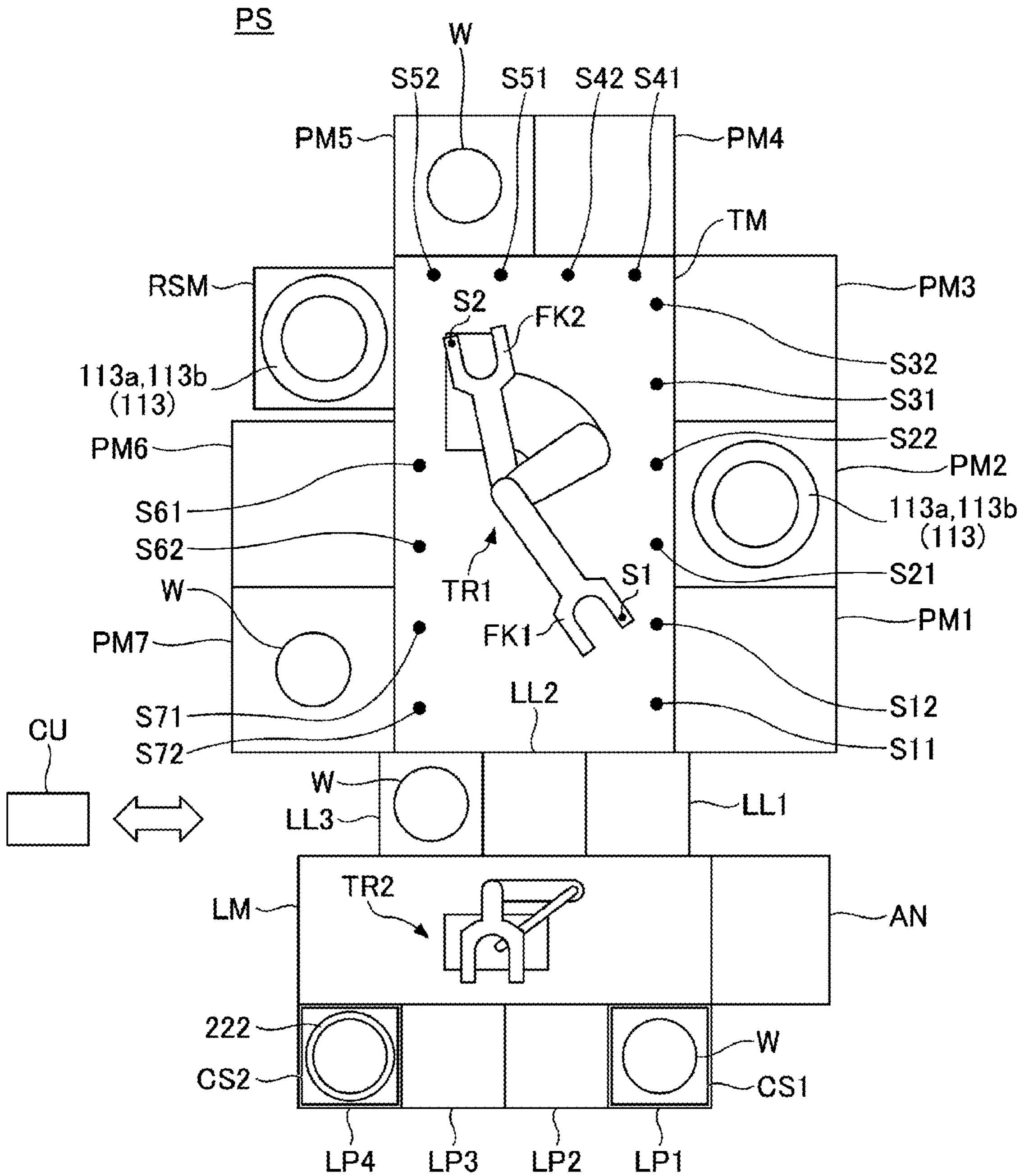

PS
W
S52
S51
S42
S41
PM5
PM4
TM
RSM
PM3
S2
FK2
S32
113a,113b
(113)
S31
PM6
S22
PM2
S61
113a,113b
(113)
S62
S21
W
TR1
S1
PM7
FK1
PM1
S71
S12
LL2
S72
S11
CU
W
LL3
LL1
TR2
LM
AN
222
W
CS2
CS1
LP4
LP3
LP2
LP1

FIG. 2

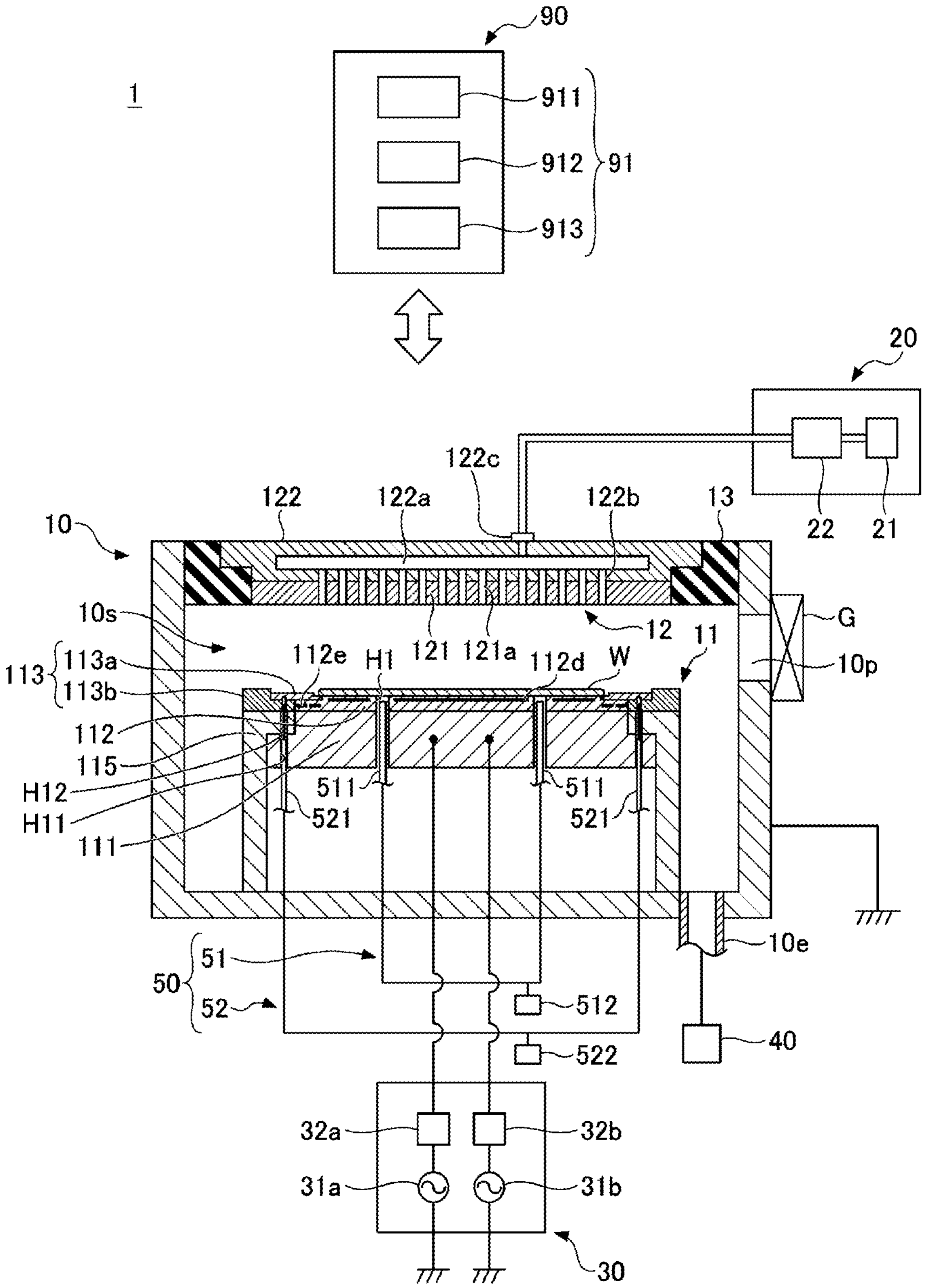
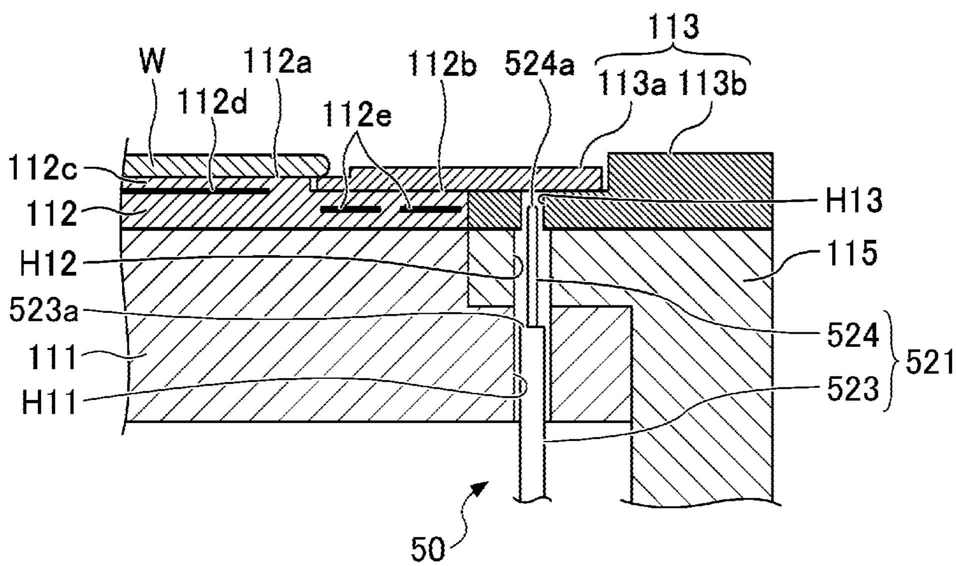
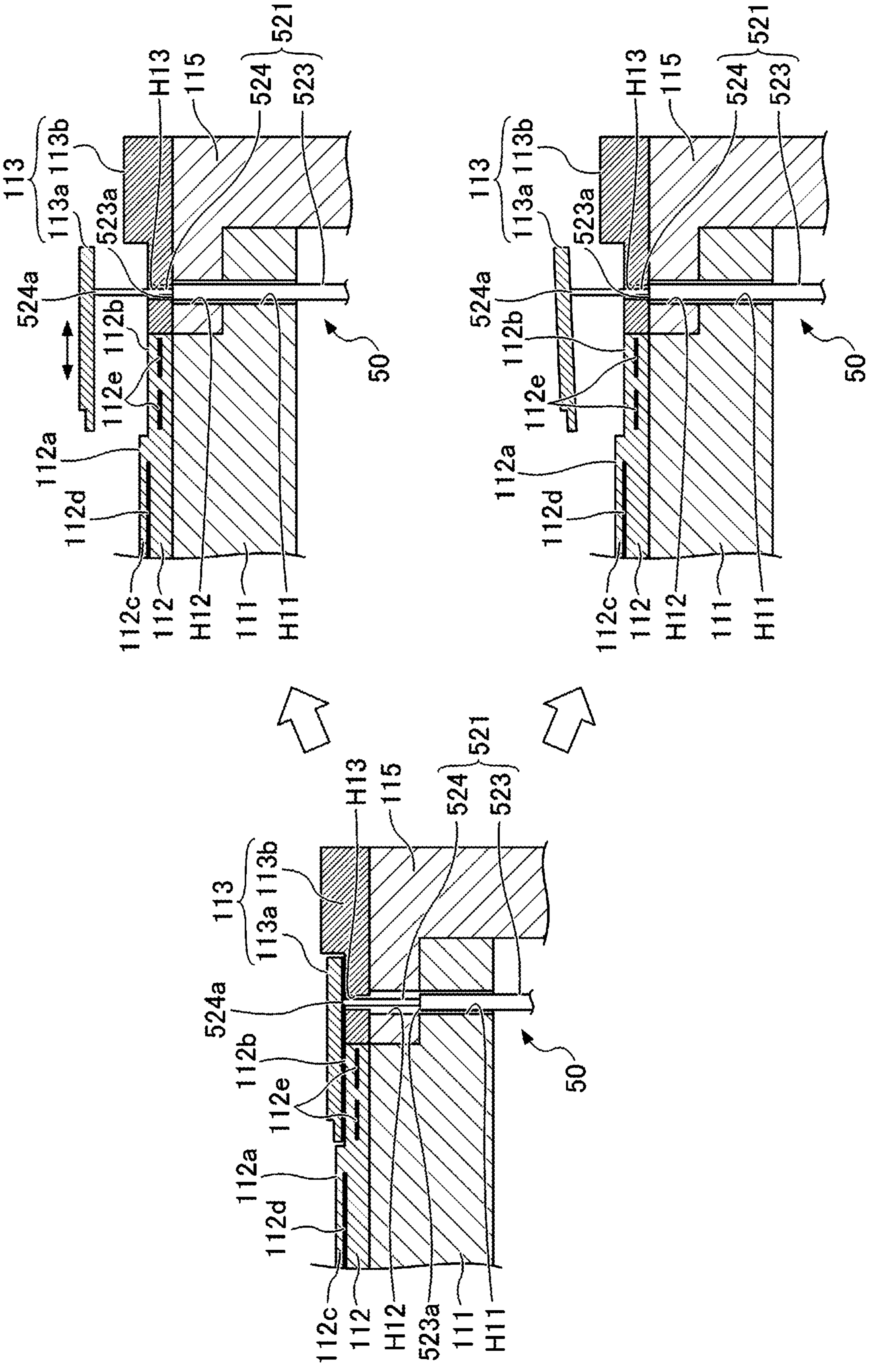

90
1
911
912
913
91
20
22
21
122c
122
122a
122b
13
10
10s
12
G
10p
113
113a
113b
112e
H1
121
121a
112d
W
11
112
115
H12
H11
111
511
511
521
521
10e
50
51
52
512
522
40
32a
32b
31a
31b
30

FIG. 3



113
W
112a
112b
524a
113a
113b
112d
112e
112c
112
H13
H12
115
523a
524
521
111
523
H11
50

FIG. 4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113a
113b
523a
524a
112b
112e
112a
112d
112c
112
H12
111
H11
H13
115
524
523
521
50
524a
112b
112e
112a
112d
112c
112
H12
523a
111
H11

FIG. 5A

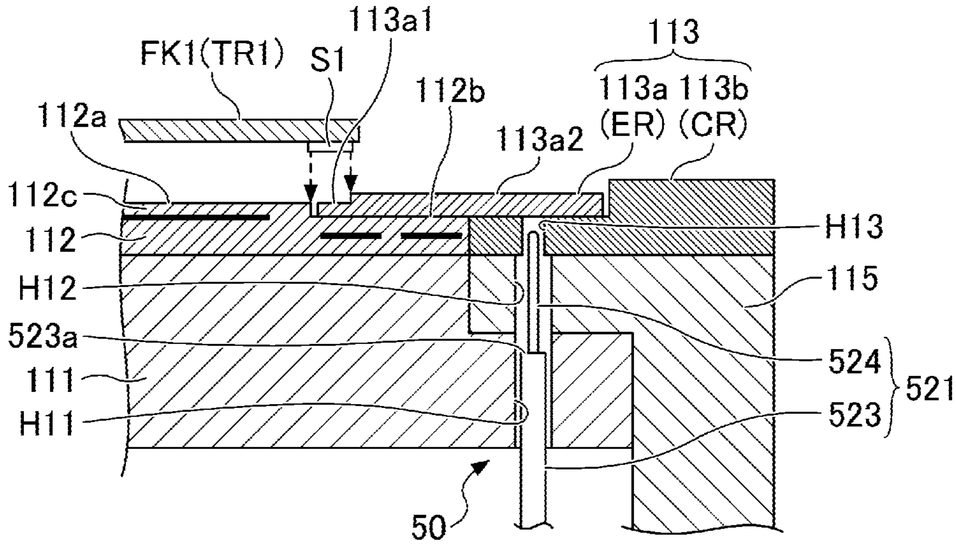
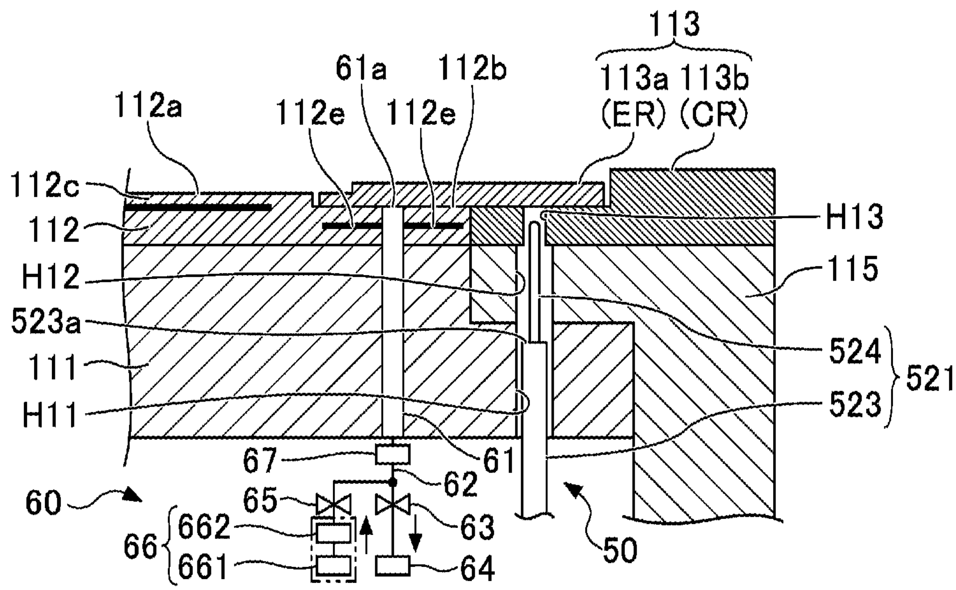

113a1
113
FK1(TR1)
S1
112b
113a 113b
112a
113a2
(ER) (CR)
112c
H13
112
H12
115
523a
524
111
521
H11
523
50

FIG. 5B


113
61a
112b
113a 113b
112a
112e
112e
(ER) (CR)
112c
H13
112
H12
115
523a
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111
521
H11
523
67
61
62
60
65
63
50
66
662
661
64

FIG. 5C

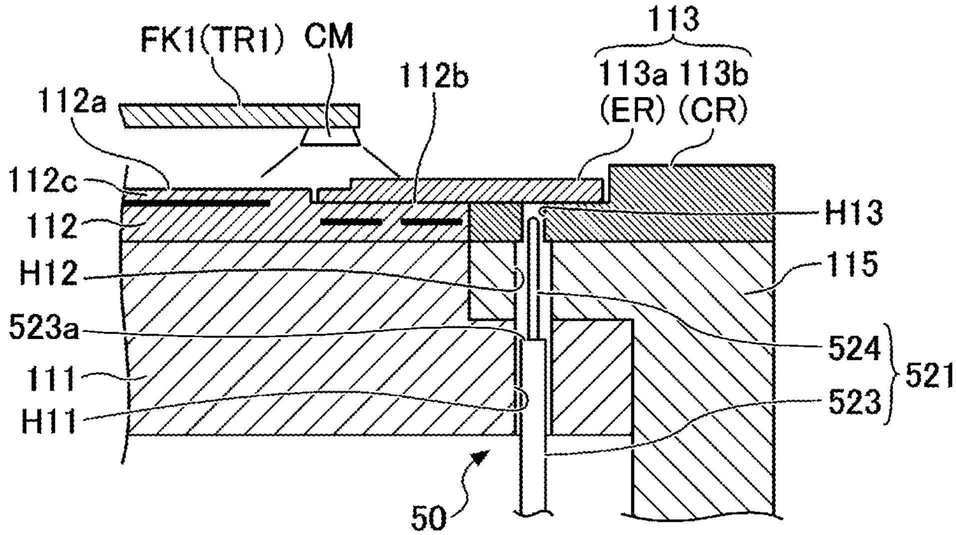

113
FK1(TR1)
CM
112b
113a
113b
(ER)
(CR)
112a
112c
112
H13
H12
115
523a
524
521
111
523
H11
50

FIG. 6

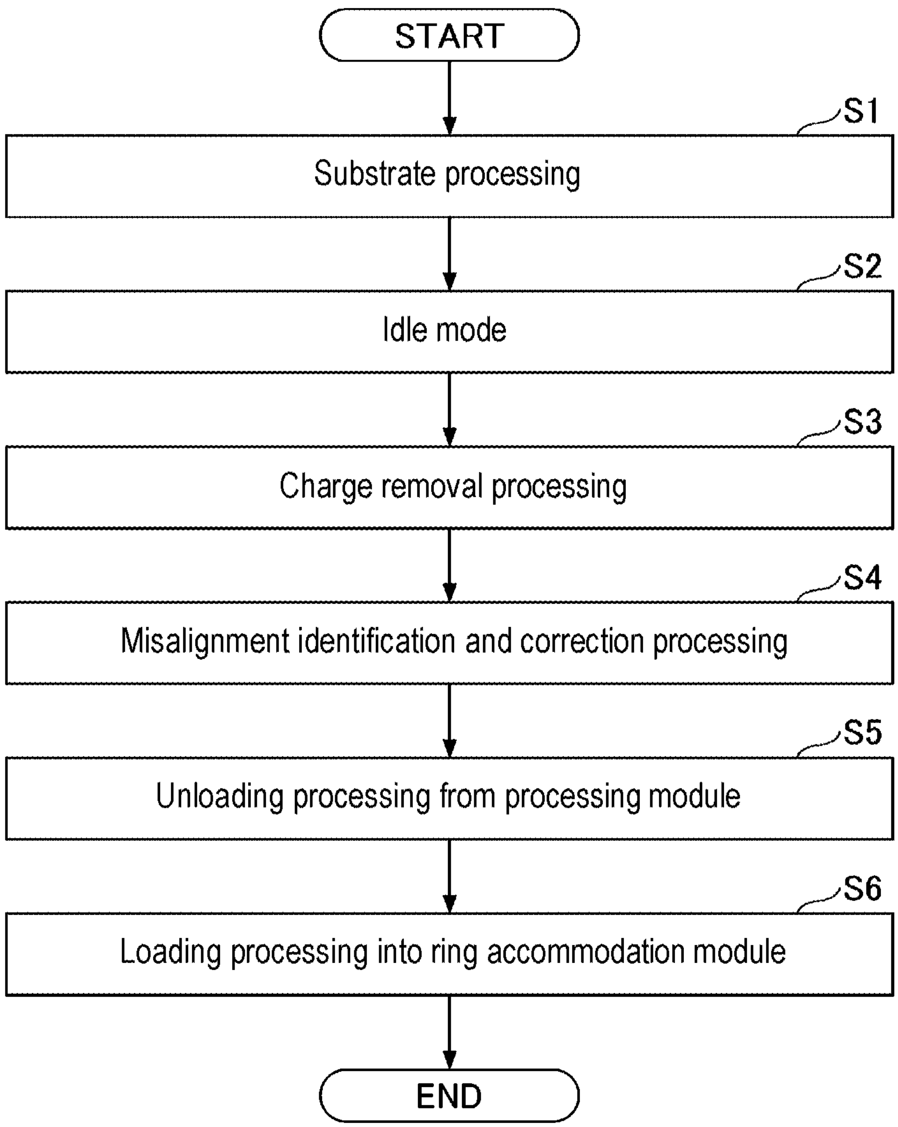
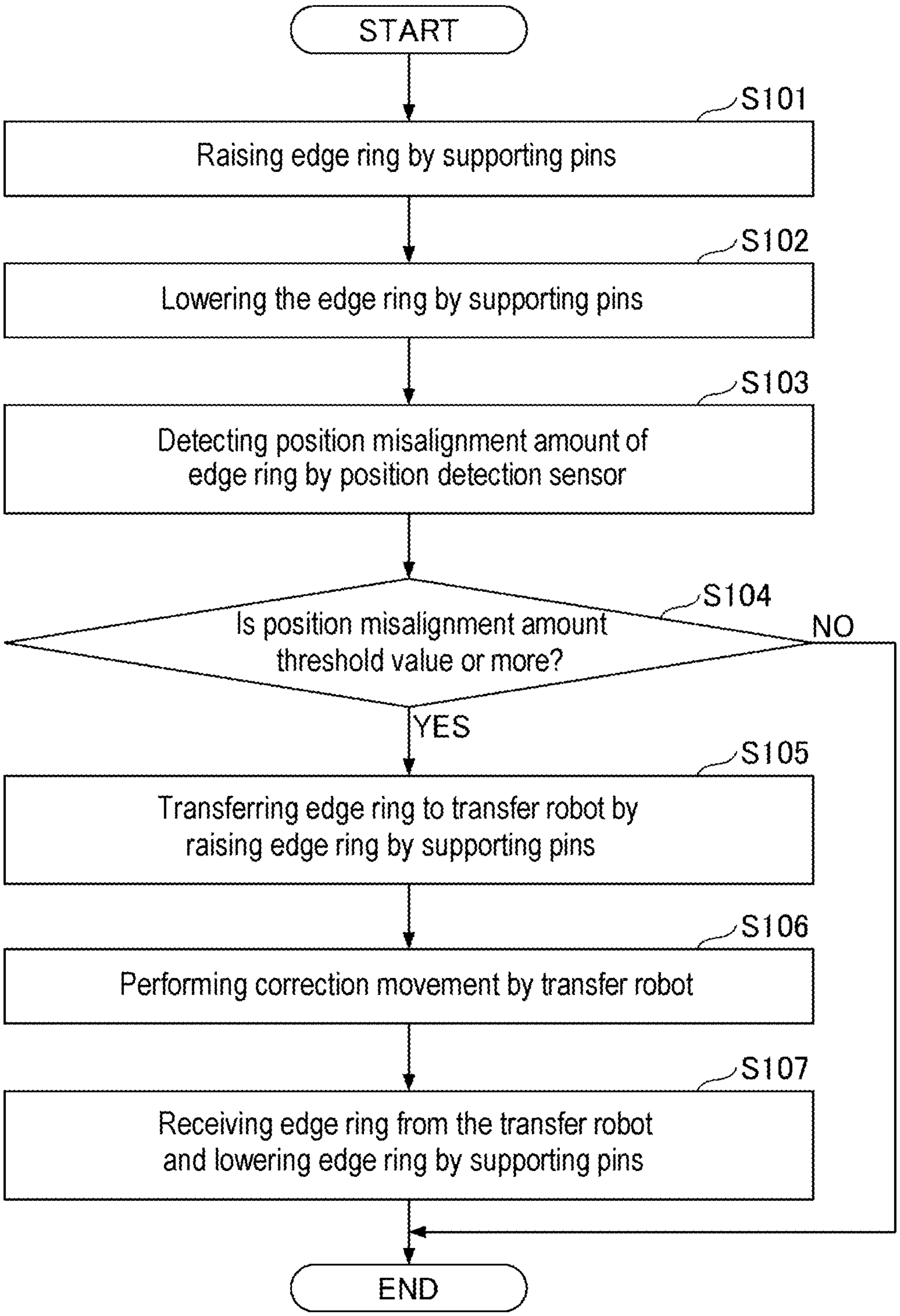

START
S1
Substrate processing
S2
Idle mode
S3
Charge removal processing
S4
Misalignment identification and correction processing
S5
Unloading processing from processing module
S6
Loading processing into ring accommodation module
END

FIG. 7


START
S101
Raising edge ring by supporting pins
S102
Lowering the edge ring by supporting pins
S103
Detecting position misalignment amount of edge ring by position detection sensor
S104
Is position misalignment amount threshold value or more?
NO
YES
S105
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S106
Performing correction movement by transfer robot
S107
Receiving edge ring from the transfer robot and lowering edge ring by supporting pins
END

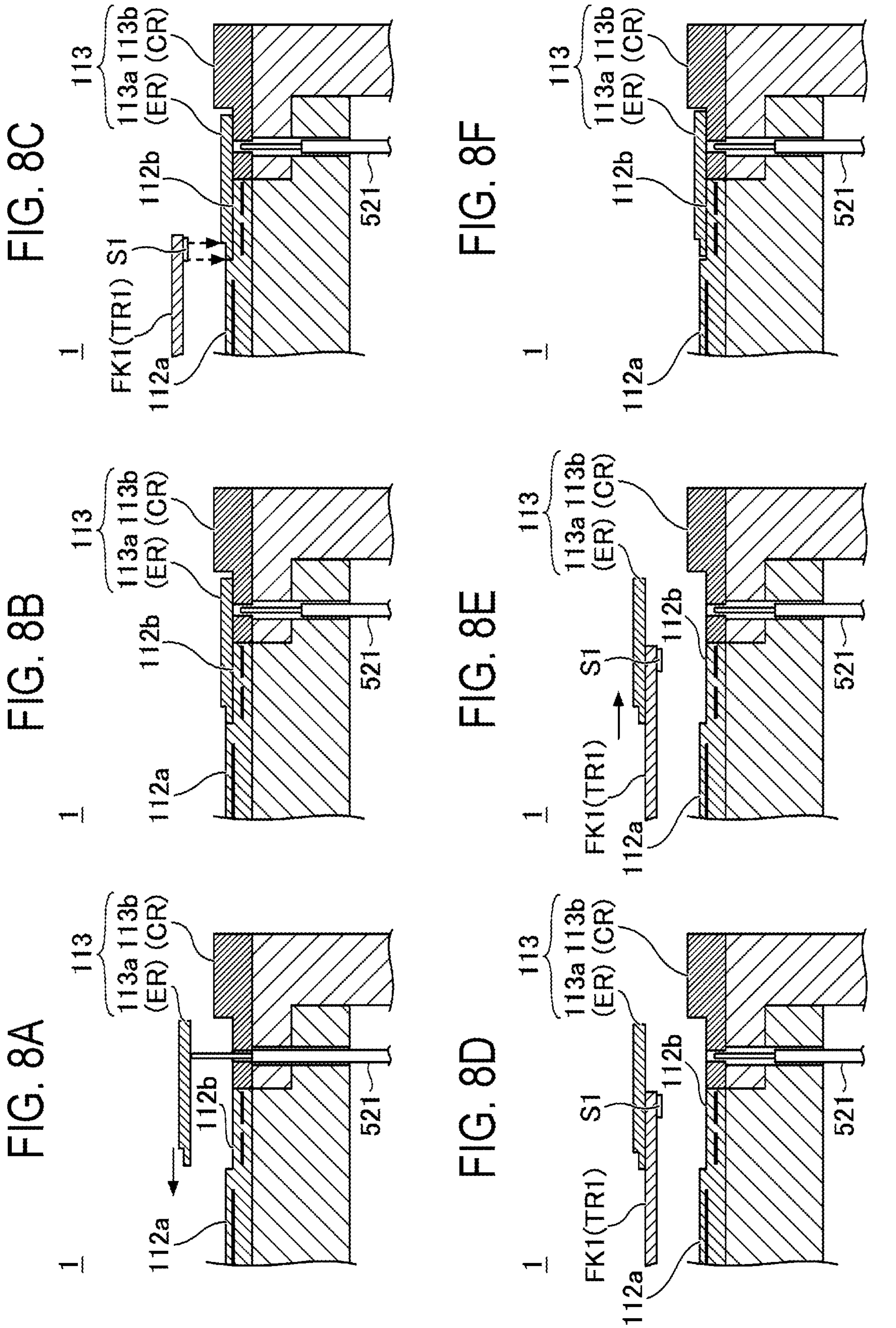


FIG. 8A

FIG. 8B

FIG. 8C

FIG. 8D

FIG. 8E

FIG. 8F

FIG. 9

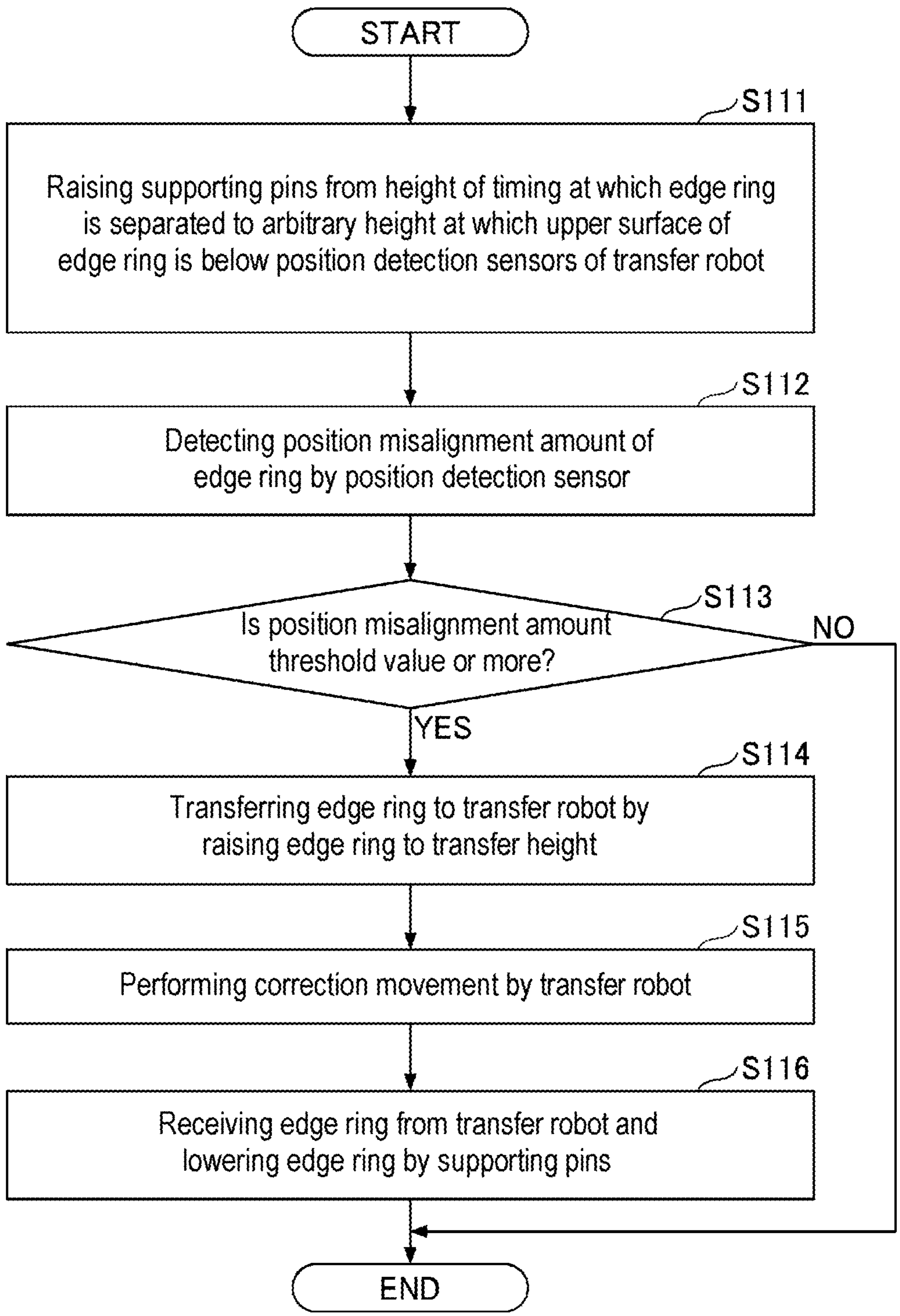

START
S111
Raising supporting pins from height of timing at which edge ring is separated to arbitrary height at which upper surface of edge ring is below position detection sensors of transfer robot
S112
Detecting position misalignment amount of edge ring by position detection sensor
S113
Is position misalignment amount threshold value or more?
NO
YES
S114
Transferring edge ring to transfer robot by raising edge ring to transfer height
S115
Performing correction movement by transfer robot
S116
Receiving edge ring from transfer robot and lowering edge ring by supporting pins
END

FIG. 10

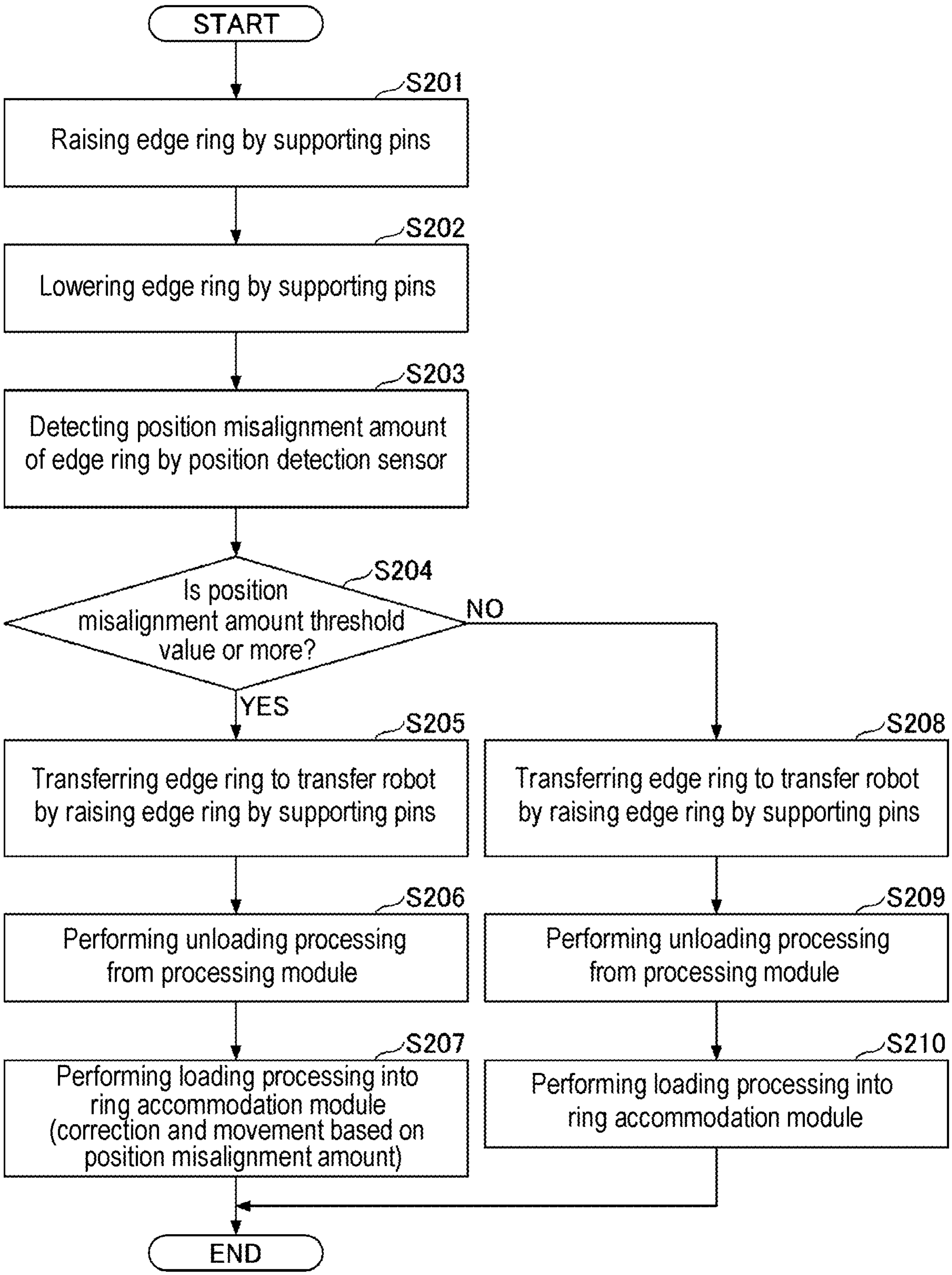
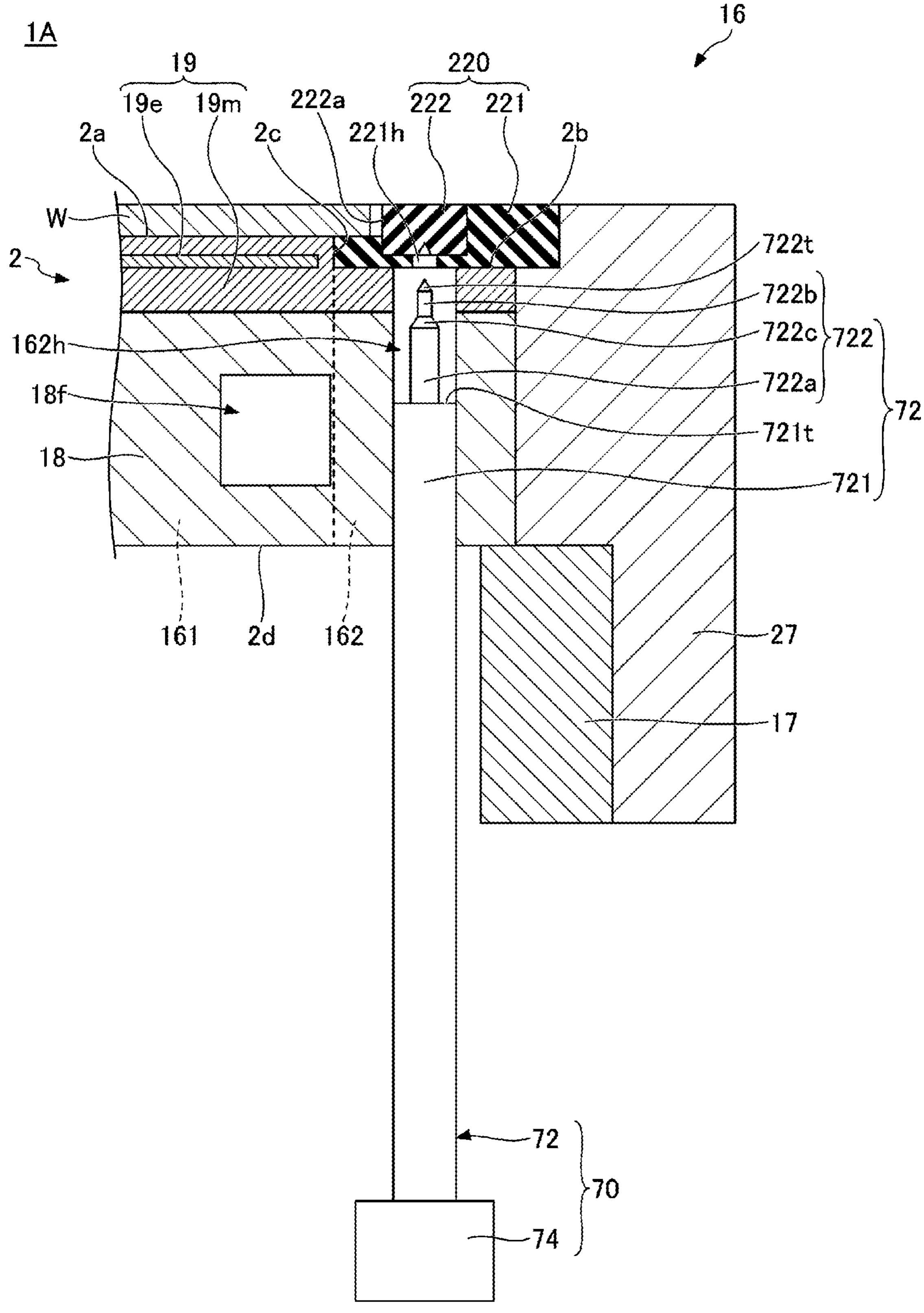

START
S201
Raising edge ring by supporting pins
S202
Lowering edge ring by supporting pins
S203
Detecting position misalignment amount of edge ring by position detection sensor
S204
Is position misalignment amount threshold value or more?
NO
YES
S205
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S206
Performing unloading processing from processing module
S207
Performing loading processing into ring accommodation module (correction and movement based on position misalignment amount)
S208
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S209
Performing unloading processing from processing module
S210
Performing loading processing into ring accommodation module
END

FIG. 11



1A
16
19
220
19e
19m
222a
222
221
2a
2c
221h
2b
W
2
722t
722b
722c
722
72
162h
722a
18f
721t
18
721
161
2d
162
27
17
72
70
74

FIG. 12

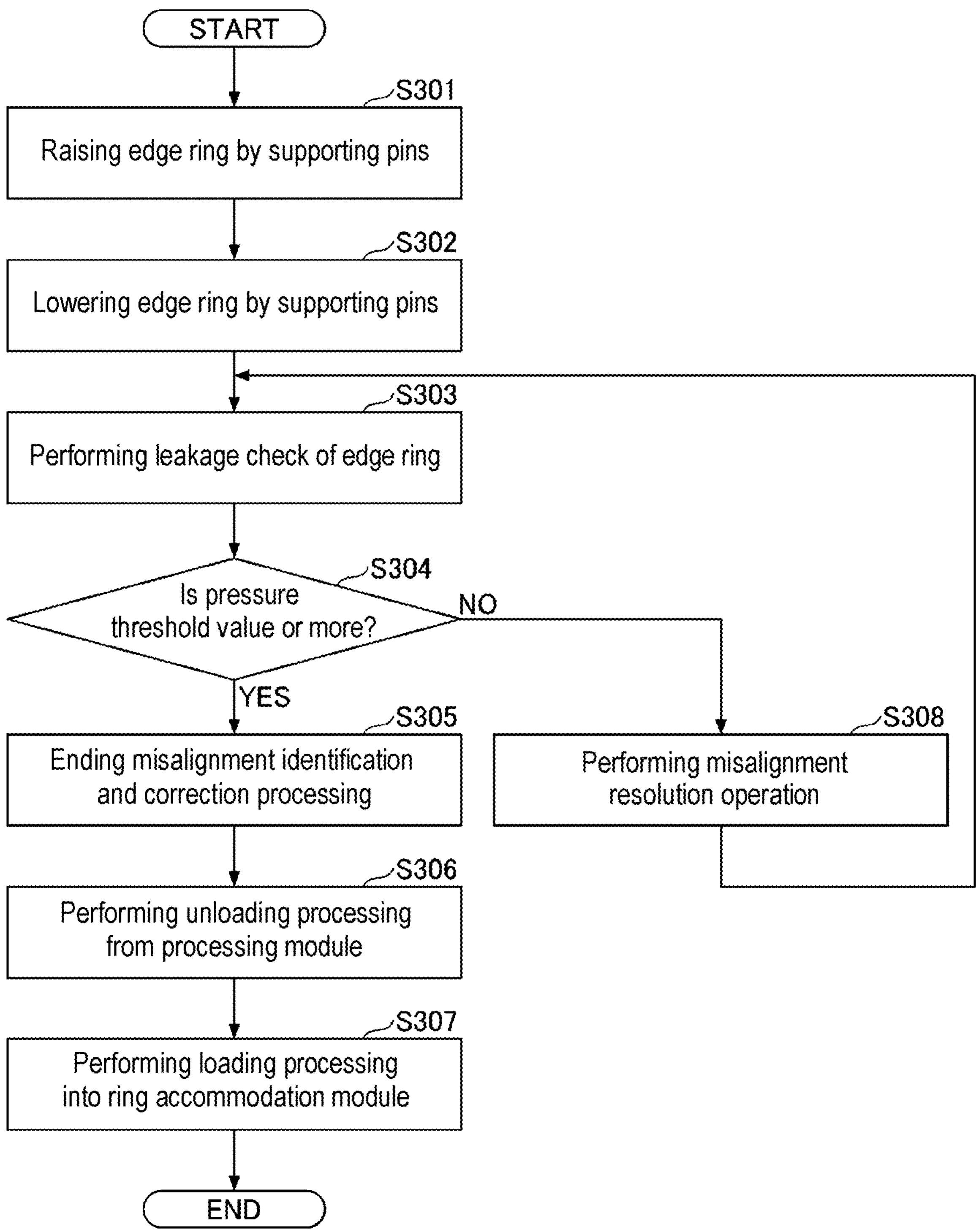

START
S301
Raising edge ring by supporting pins
S302
Lowering edge ring by supporting pins
S303
Performing leakage check of edge ring
S304
Is pressure threshold value or more?
NO
YES
S305
Ending misalignment identification and correction processing
S308
Performing misalignment resolution operation
S306
Performing unloading processing from processing module
S307
Performing loading processing into ring accommodation module
END

FIG. 13C

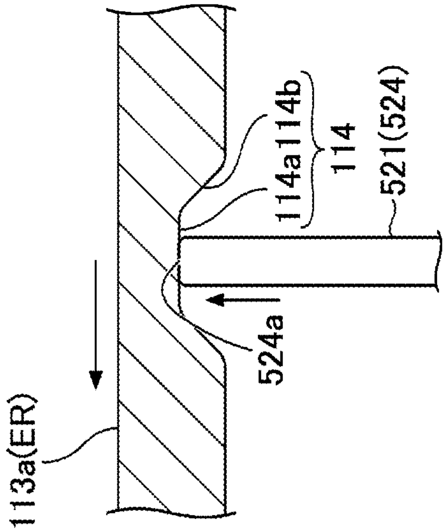


FIG. 13B

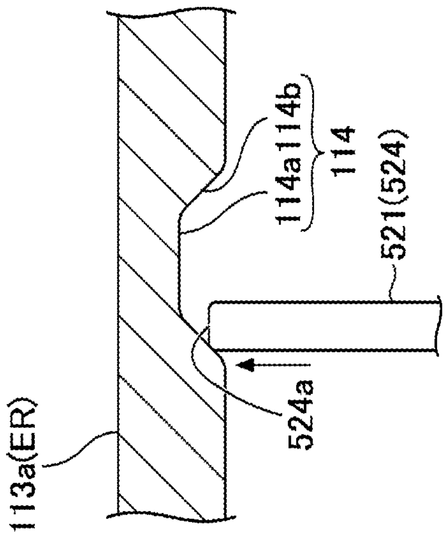


FIG. 13A

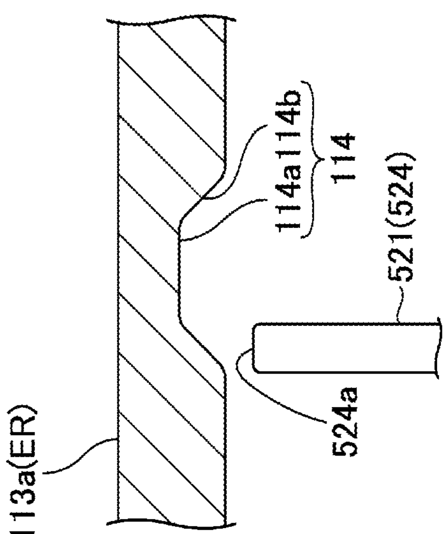


FIG. 14

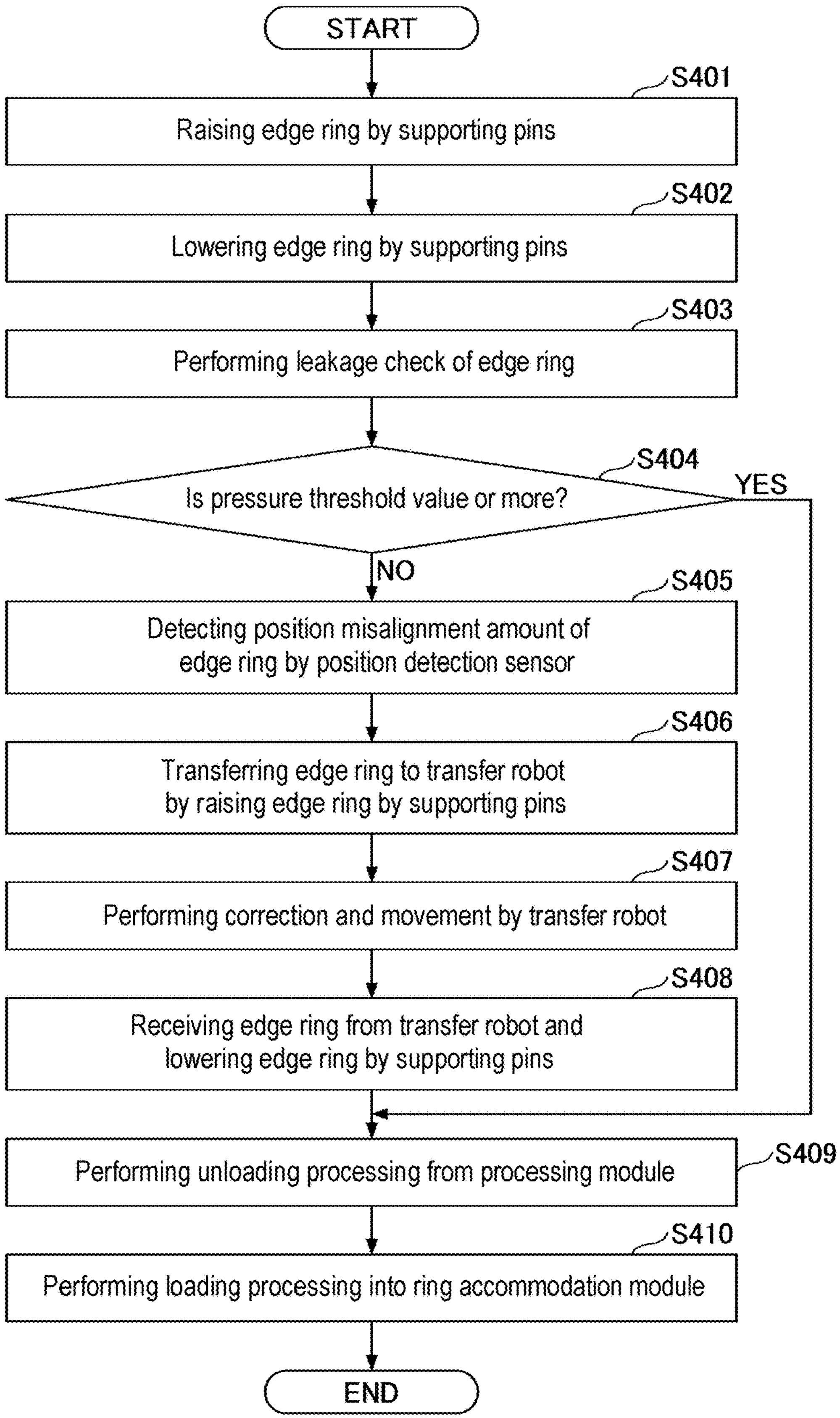

START
S401
Raising edge ring by supporting pins
S402
Lowering edge ring by supporting pins
S403
Performing leakage check of edge ring
S404
Is pressure threshold value or more?
YES
NO
S405
Detecting position misalignment amount of edge ring by position detection sensor
S406
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S407
Performing correction and movement by transfer robot
S408
Receiving edge ring from transfer robot and lowering edge ring by supporting pins
S409
Performing unloading processing from processing module
S410
Performing loading processing into ring accommodation module
END

FIG. 15

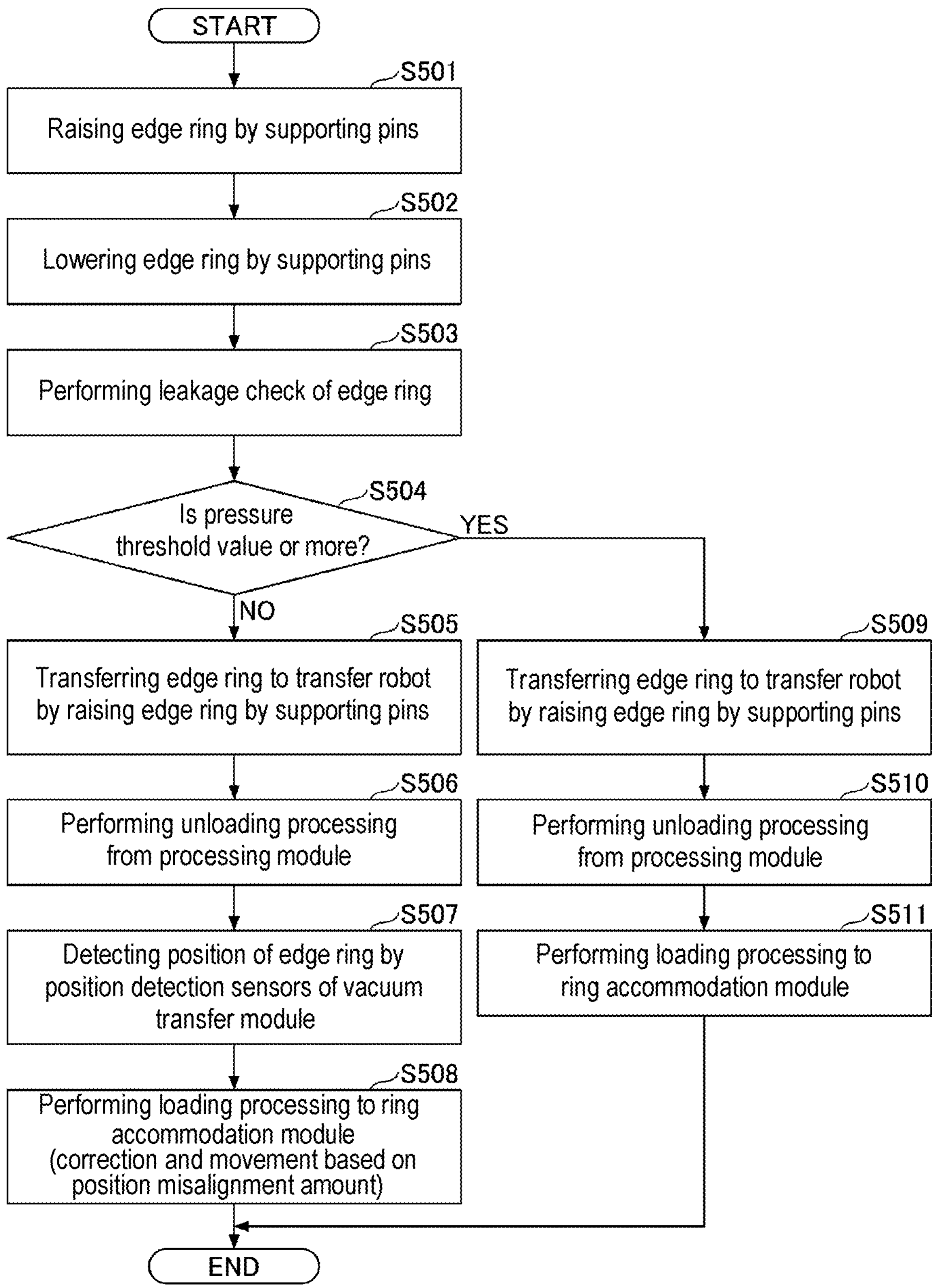

START
S501
Raising edge ring by supporting pins
S502
Lowering edge ring by supporting pins
S503
Performing leakage check of edge ring
S504
Is pressure threshold value or more?
YES
NO
S505
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S506
Performing unloading processing from processing module
S507
Detecting position of edge ring by position detection sensors of vacuum transfer module
S508
Performing loading processing to ring accommodation module (correction and movement based on position misalignment amount)
S509
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S510
Performing unloading processing from processing module
S511
Performing loading processing to ring accommodation module
END

FIG. 16

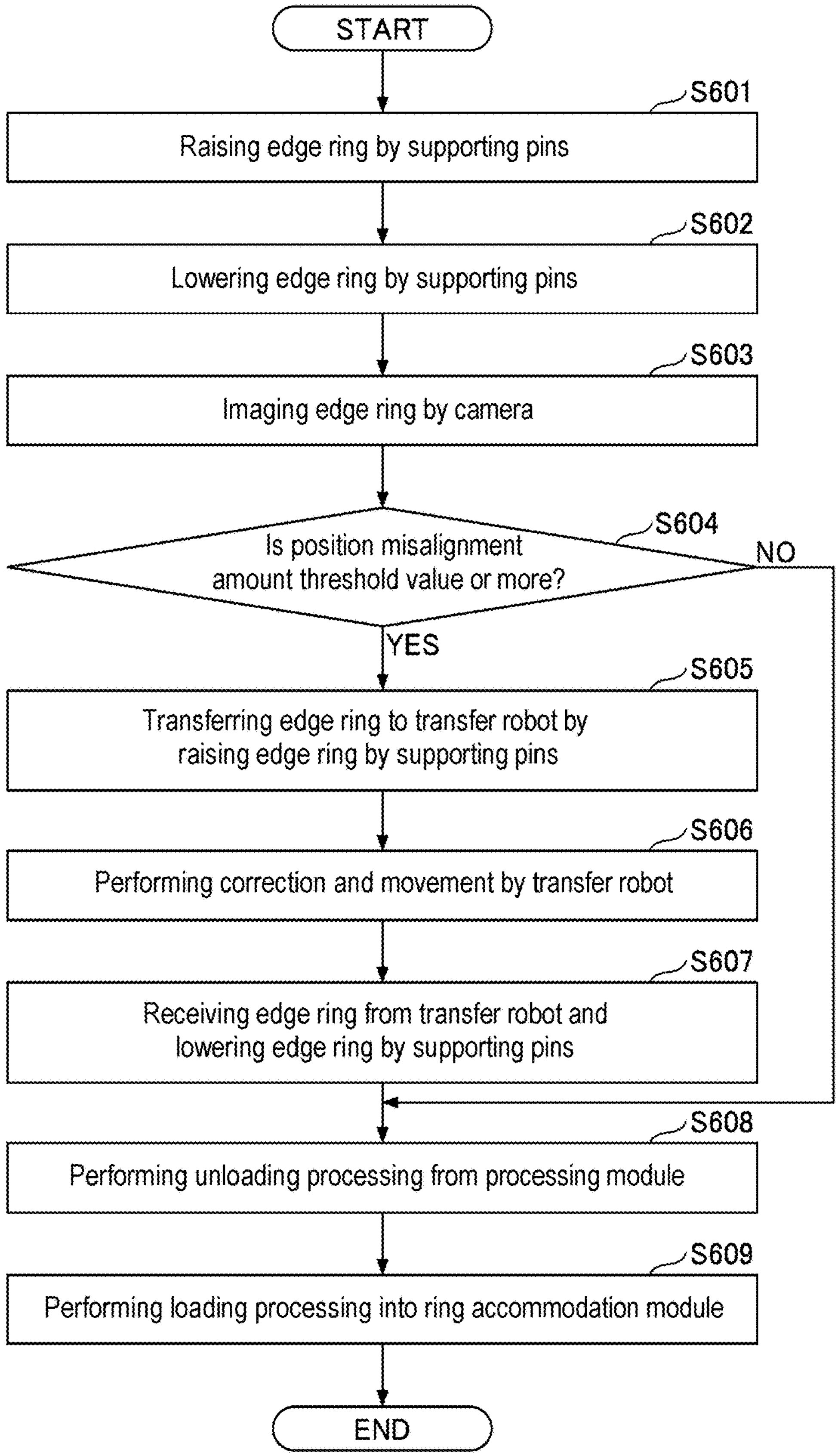

START
S601
Raising edge ring by supporting pins
S602
Lowering edge ring by supporting pins
S603
Imaging edge ring by camera
S604
Is position misalignment amount threshold value or more?
NO
YES
S605
Transferring edge ring to transfer robot by raising edge ring by supporting pins
S606
Performing correction and movement by transfer robot
S607
Receiving edge ring from transfer robot and lowering edge ring by supporting pins
S608
Performing unloading processing from processing module
S609
Performing loading processing into ring accommodation module
END

FIG. 17

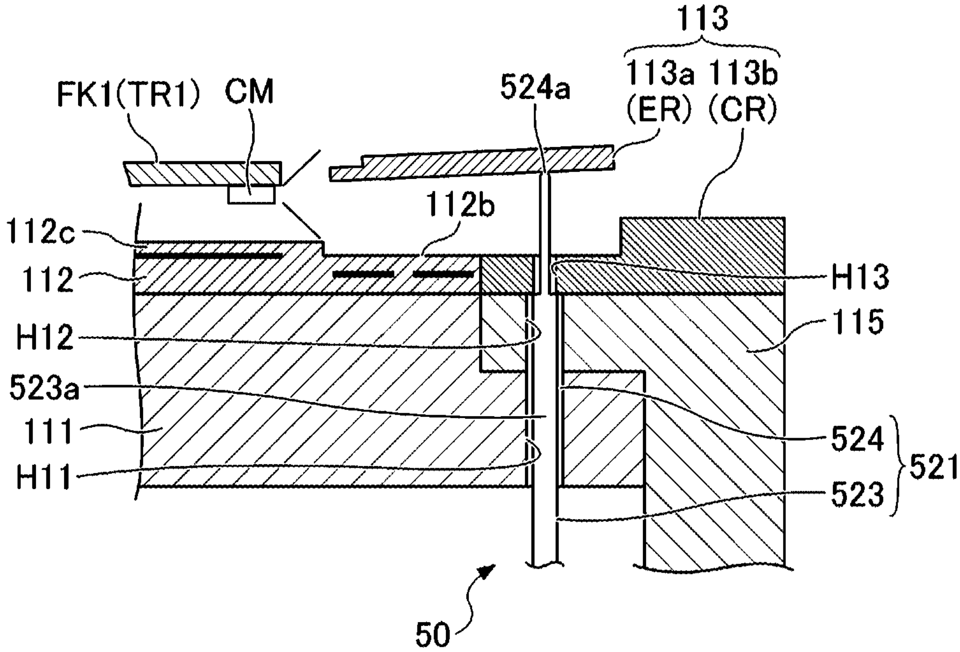

113
FK1(TR1)
CM
524a
113a 113b
(ER) (CR)
112c
112b
112
H13
H12
115
523a
111
524
521
H11
523
50

FIG. 18

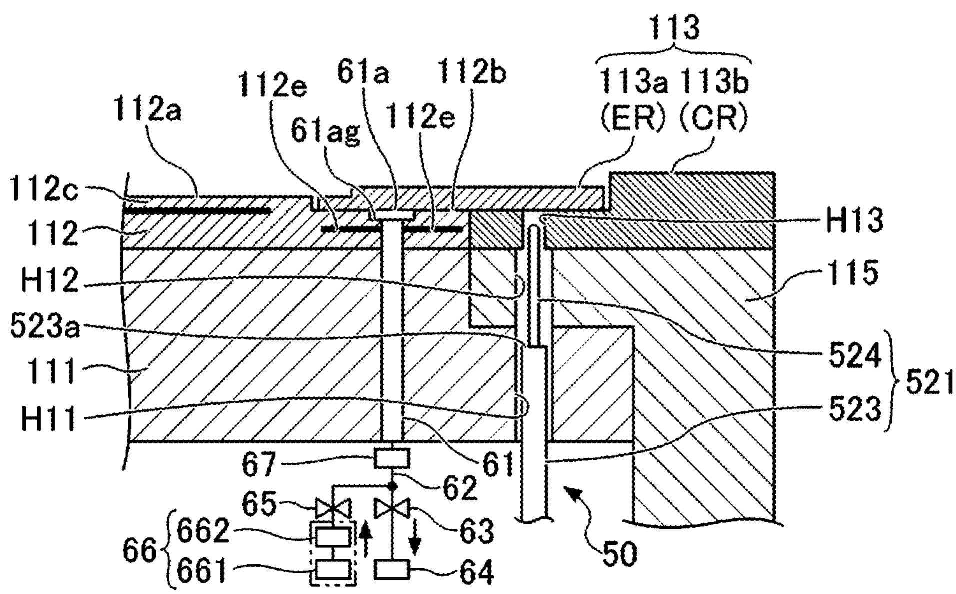

113
112a
112e
61a
112b
113a 113b
61ag
112e
(ER) (CR)
112c
112
H13
H12
115
523a
524
521
111
H11
523
67
61
62
65
63
66
662
661
64
50

FIG. 19

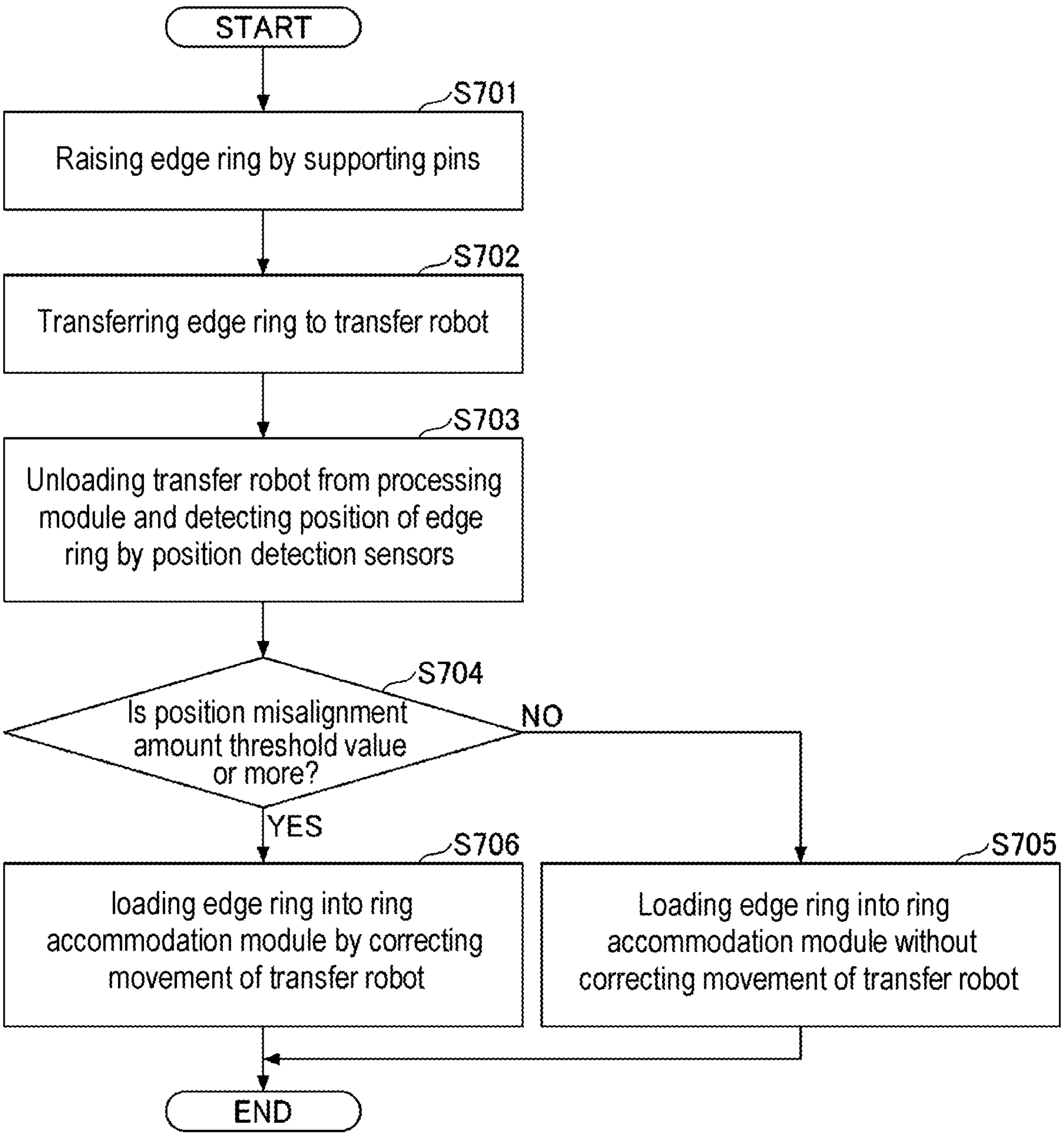

START
S701
Raising edge ring by supporting pins
S702
Transferring edge ring to transfer robot
S703
Unloading transfer robot from processing module and detecting position of edge ring by position detection sensors
S704
Is position misalignment amount threshold value or more?
NO
YES
S706
loading edge ring into ring accommodation module by correcting movement of transfer robot
S705
Loading edge ring into ring accommodation module without correcting movement of transfer robot
END

FIG. 20A

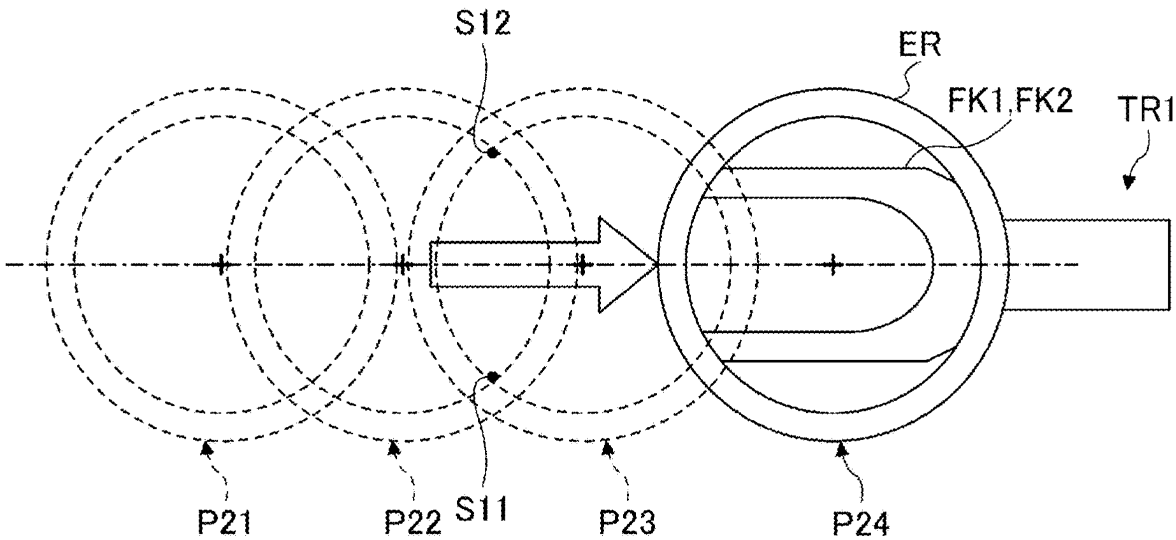
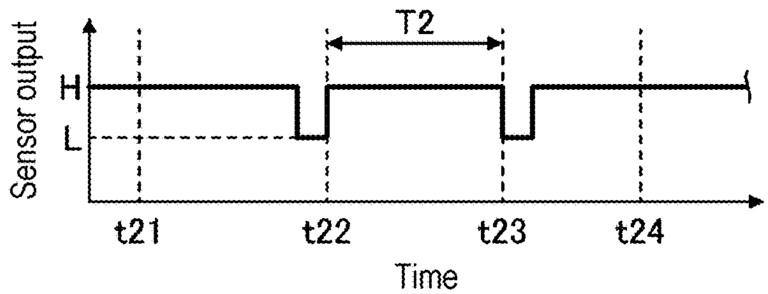


FIG. 20B



SUBSTRATE PROCESSING SYSTEM AND TRANSFER METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The application is a Bypass Continuation application of PCT International Application No. PCT/JP2023/038477, filed on Oct. 25, 2023 and designating the United States, the international application being based upon and claiming the benefit of priority from Japanese Patent Application No. 2022-174836, filed on Oct. 31, 2022, the entire content of each are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a substrate processing system and a transfer method.

BACKGROUND

[0003] Patent Document 1 discloses a plasma processing apparatus which performs a plasma processing on a substrate placed on a stage (substrate support) provided inside a processing chamber by disposing a focus ring (ring) at a periphery of the substrate support. A substrate processing system including the plasma processing apparatus performs an operation of, when exchanging the ring, taking out the ring from the substrate support by a transfer robot, cleaning a surface on which the ring is placed, and placing a ring on the substrate support again.

PRIOR ART DOCUMENTS

Patent Documents

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. 2018-010992

SUMMARY

[0005] According to one embodiment of the present disclosure, there is provided a substrate processing system including: a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring; a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and a controller. The controller performs: (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support; (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

BRIEF DESCRIPTION OF DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.

[0007] FIG. **1** is a view illustrating an example of a substrate processing system according to an embodiment.

[0008] FIG. **2** is a schematic cross-sectional view illustrating an example of a plasma processing apparatus.

[0009] FIG. **3** is an enlarged view of a portion in FIG. **2**.

[0010] FIG. **4** is a view illustrating a cause of a position misalignment which occurs in a ring when taking out the ring from a substrate support.

[0011] FIG. **5**A is a view illustrating a first example of detecting an index related to a position misalignment amount of an inner ring. FIG. **5**B is a view illustrating a second example of detecting an index related to a position misalignment amount of the inner ring. FIG. **5**C is a view illustrating a third example of detecting an index related to a position misalignment amount of the inner ring.

[0012] FIG. **6** is a flowchart illustrating an operation sequence when taking out an edge ring.

[0013] FIG. **7** is a flowchart illustrating a processing flow of a misalignment identification and correction processing in a transfer method according to a first example.

[0014] FIG. **8**A is a first view illustrating an operation of the misalignment identification and correction processing. FIG. **8**B is a second view illustrating an operation continued from FIG. **8**A. FIG. **8**C is a third view illustrating an operation continued from FIG. **8**B. FIG. **8**D is a fourth view illustrating an operation continued from FIG. **8**C. FIG. **8**E is a fifth view illustrating an operation continued from FIG. **8**D. FIG. **8**F is a sixth view illustrating an operation continued from FIG. **8**E.

[0015] FIG. **9** is a flowchart illustrating a modification of the processing flow of the misalignment identification and correction processing according to the first example.

[0016] FIG. **10** is a flowchart illustrating a processing flow in a transfer method according to a second example.

[0017] FIG. **11** is a schematic cross-sectional view illustrating another example of the plasma processing apparatus.

[0018] FIG. **12** is a flowchart illustrating a processing flow in a transfer method according to a third example.

[0019] FIG. **13**A is a first view illustrating a misalignment resolution operation. FIG. **13**B is a second view illustrating an operation continued from FIG. **13**A. FIG. **13**C is a third view illustrating an operation continued from FIG. **13**B.

[0020] FIG. **14** is a flowchart illustrating a processing flow in a transfer method according to a fourth example.

[0021] FIG. **15** is a flowchart illustrating a processing flow in a transfer method according to a fifth example.

[0022] FIG. **16** is a flowchart illustrating a processing flow in a transfer method according to a sixth example.

[0023] FIG. **17** is a view illustrating an example of imaging an inclined edge ring by a camera.

[0024] FIG. **18** is a view illustrating a modification of a configuration which performs a leakage check of gas as an index related to a position misalignment amount of the edge ring.

[0025] FIG. **19** is a flowchart illustrating a transfer operation sequence of the edge ring according to a modification.

[0026] FIG. **20**A is a view illustrating a relationship between a position of the edge ring and positions of position detection sensors. FIG. **20**B is a view illustrating a change in sensor output of the position detection sensors when transferring the edge ring from one position to the other position.

DETAILED DESCRIPTION

[0027] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0028] Hereinafter, embodiments for implementing the present disclosure will be described with reference to the drawings. In the respective drawings, the same components may be denoted by the same reference numerals, and duplicated descriptions thereof may be omitted.

[Substrate Processing System]

[0029] A substrate processing system PS according to an embodiment is described with reference to FIG. **1**. FIG. **1** is a view illustrating an example of the substrate processing system PS according to the embodiment. As illustrated in FIG. **1**, the substrate processing system PS is a system capable of performing various types of processings, such as a plasma processing, on a substrate W. The substrate W may be, for example, a semiconductor wafer.

[0030] The substrate processing system PS includes a vacuum transfer module TM, a plurality of processing modules PM**1** to PM**7**, a ring accommodation module RSM, a plurality of load lock modules LL**1** to LL**3**, an atmospheric transfer module LM, load ports LP**1** to LP**4**, an aligner AN, and a controller CU. The vacuum transfer module TM is also referred to as a transfer module. The processing modules PM**1** to PM**7** are also referred to as process modules. The ring accommodation module RSM is also referred to as a ring stocker module. The atmospheric transfer module LM is also referred to as a loader module.

[0031] The vacuum transfer module TM has a quadrangular shape in a plan view. The processing modules PM**1** to PM**7**, the load lock modules LL**1** to LL**3**, and the ring accommodation module RSM are connected to the vacuum transfer module TM. The vacuum transfer module TM includes a vacuum transfer chamber. The vacuum transfer chamber is maintained in a vacuum atmosphere. A transfer robot TR**1** is installed in the vacuum transfer chamber (inside the vacuum transfer module TM).

[0032] The transfer robot TR**1** is configured to be rotatable, extensible, and movable vertically. The transfer robot TR**1** includes an upper fork FK**1** and a lower fork FK**2**. The upper fork FK**1** and the lower fork FK**2** of the transfer robot TR**1** are configured to be capable of holding each of the substrate W and a ring **113** (an inner ring **113***a* and an outer ring **113***b*). The transfer robot TR**1** holds and transfers the substrate W and the ring **113** among the processing modules PM**1** to PM**7**, the load lock modules LL**1** to LL**3**, and the ring accommodation module RSM.

[0033] The upper fork FK**1** is provided with a position detection sensor S**1**. The lower fork FK**2** is provided with a position detection sensor S**2**. The position detection sensors S**1** and S**2** detect positions of the inner ring **113***a* and the outer ring **113***b*, which are loaded in the processing modules PM**1** to PM**7**. The position detection sensors S**1** and S**2** may be, for example, optical displacement sensors, cameras or the like.

[0034] The vacuum transfer module TM may be provided with position detection sensors S**11** and S**12**. The position detection sensors S**11** and S**12** are installed on a transfer path of the substrate W and the ring **113** (the inner ring **113***a*), which are transferred from the vacuum transfer module TM to the processing module PM**1**. The position detection sensors S**11** and S**12** are used when loading the substrate W or the ring **113** from the vacuum transfer module TM to the processing module PM**1** and when unloading the substrate W or the ring **113** from the processing module PM**1** to the vacuum transfer module TM. The position detection sensors S**11** and S**12** are installed, for example, in a vicinity of a gate valve (not illustrated) which partitions the vacuum transfer module TM and the processing module PM**1**. The position detection sensors S**11** and S**12** are disposed, for example, such that a distance between the position detection sensors S**11** and S**12** is smaller than an outer diameter of the substrate W and is smaller than an inner diameter of the inner ring **113***a*. Like the position detection sensors S**11** and S**12**, the vacuum transfer module TM may be provided with position detection sensors S**21**, S**22**, S**31**, S**32**, S**41**, S**42**, S**51**, S**52**, S**61**, S**62**, S**71**, and S**72**.

[0035] The processing modules PM**1** to PM**7** are connected to the vacuum transfer module TM. The processing modules PM**1** to PM**7** have vacuum processing chambers. A substrate support **11** (see FIG. **2**) is installed inside the vacuum processing chamber. After the substrate W is placed on the substrate support **11**, the processing modules PM**1** to PM**7** reduce the internal pressure, introduce a processing gas, and apply RF power applied to generate plasma, whereby a plasma processing is performed on the substrate W by the plasma. The vacuum transfer module TM is partitioned from the processing modules PM**1** to PM**7** by openable/closable gate valves (not illustrated).

[0036] The ring accommodation module RSM is an example of an apparatus which accommodates the ring **113**, and is connected to the vacuum transfer module TM. The ring accommodation module RSM accommodates, for example, the inner ring **113***a* and the outer ring **113***b*, which constitute the ring **113**. The ring accommodation module RSM may be configured to accommodate only the inner ring **113***a*. The ring accommodation module RSM may be configured to accommodate only the outer ring **113***b*. The inner ring **113***a* and the outer ring **113***b* are transferred between the processing modules PM**1** to PM**7** and the ring accommodation module RSM by the transfer robot TR**1**. The vacuum transfer module TM and the ring accommodation module RSM are partitioned by an openable/closable gate valve (not illustrated).

[0037] The load lock modules LL**1** to LL**3** are installed between the vacuum transfer module TM and the atmospheric transfer module LM. The load lock modules LL**1** to LL**3** are connected to the vacuum transfer module TM and the atmospheric transfer module LM. Each of the load lock modules LL**1** to LL**3** includes therein an internal pressure variable chamber of which an inside is switchable between vacuum and atmospheric pressure. The internal pressure variable chamber is provided with a stage (not illustrated) which enables the substrate W to be placed thereon. When transferring the substrate W from the atmospheric transfer module LM to the vacuum transfer module TM, the load lock modules LL**1** to LL**3** receive the substrate W from the atmospheric transfer module LM while maintaining the inside of the internal pressure variable chamber at the atmospheric pressure, and then depressurize the inside of the internal pressure variable chamber to transfer the substrate

W to the vacuum transfer module TM. When transferring the substrate W from the vacuum transfer module TM to the atmospheric transfer module LM, the load lock modules LL**1** to LL**3** receive the substrate W from the vacuum transfer module TM while maintaining the inside of the internal pressure variable chamber in the vacuum, and then pressurize the inside of the internal pressure variable chamber to the atmospheric pressure to transfer the substrate W to the atmospheric transfer module LM. The load lock modules LL**1** to LL**3** are partitioned from the vacuum transfer module TM by openable/closable gate valves (not illustrated). The load lock modules LL**1** to LL**3** are partitioned from the atmospheric transfer module LM by openable/closable gate valves (not illustrated).

[0038] The atmospheric transfer module LM is installed to face the vacuum transfer module TM. The atmospheric transfer module LM may be, for example, an equipment front end module (EFEM). The atmospheric transfer module LM has a quadrangular shape in a plan view. The atmospheric transfer module LM includes an atmospheric transfer chamber. An inside of the atmospheric transfer chamber is maintained in an atmospheric pressure atmosphere. A transfer robot TR**2** is installed inside the atmospheric transfer chamber. The transfer robot TR**2** holds and transfers the substrate W among the load ports LP**1** to LP**4**, the aligner AN, and the load lock modules LL**1** to LL**3**. The atmospheric transfer module LM may have a fan filter unit (FFU).

[0039] The load ports LP**1** to LP**4** are connected to the atmospheric transfer module LM. A plurality of substrate accommodation containers CS**1** is placed in the load lock ports LP**1** to LP**4**. The substrate accommodation container CS**1** may be, for example, a front-opening unified pod (FOUP) which accommodates a plurality of substrates W (for example, 25 substrates W).

[0040] The aligner AN is connected to the atmospheric transfer module LM. The aligner AN is configured to perform position adjustment of the substrate W. The aligner AN may be installed inside the atmospheric transfer chamber.

[0041] The controller CU controls each part of the substrate processing system PS. The controller CU controls, for example, an operation of the transfer robot TR**1** installed in the vacuum transfer module TM, an operation of the transfer robot TR**2** installed in the atmospheric transfer module LM, and opening and closing of the gate valves. The controller CU may be, for example, a computer. The controller CU includes a central processing unit (CPU) which is a processor, a random access memory (RAM), a read only memory (ROM), an auxiliary storage device, and the like. The CPU operates based on a program stored in the ROM or the auxiliary storage device, and controls each part of the substrate processing system PS.

[Plasma Processing Apparatus]

[0042] An example of a plasma processing apparatus **1** applied to the processing modules PM**1** to PM**7** in FIG. **1** is described with reference to FIGS. **2** and **3**. FIG. **2** is a schematic cross-sectional view illustrating an example of the plasma processing apparatus **1**. FIG. **3** is an enlarged view of a portion in FIG. **2**.

[0043] The plasma processing apparatus **1** includes a plasma processing chamber **10** (processing chamber), a gas supply **20**, an RF power supply **30**, an exhaust system **40**, a lifter **50**, and a controller **90**.

[0044] The plasma processing chamber **10** includes a substrate support **11** and an upper electrode **12**. The substrate support **11** is disposed in a lower region of a plasma processing space **10***s* in the plasma processing chamber **10**. The upper electrode **12** is disposed above the substrate support **11**, to function as a portion of a ceiling plate of the plasma processing chamber **10**.

[0045] The substrate support **11** supports the substrate W in the plasma processing space **10***s*. The substrate support **11** includes a lower electrode **111**, an electrostatic chuck **112**, a ring **113** (hereinafter, also referred to as a ring assembly **113**), and an insulating member **115**.

[0046] The electrostatic chuck **112** is disposed on the lower electrode **111**. The electrostatic chuck **112** has an upper surface which includes a substrate support surface **112***a* and a ring support surface **112***b*. The electrostatic chuck **112** supports the substrate W on the substrate support surface **112***a*. The electrostatic chuck **112** supports the inner ring **113***a* on the ring support surface **112***b*. The electrostatic chuck **112** includes an insulating member **112***c*, a first attraction electrode **112***d*, and a second attraction electrode **112***e*. The first attraction electrode **112***d* and the second attraction electrode **112***e* are embedded in the insulating member **112***c*. The first attraction electrode **112***d* is located below the substrate support surface **112***a*. The electrostatic chuck **112** attracts and holds the substrate W on the substrate support surface **112***a* by applying a voltage to the first attraction electrode **112***d*. The second attraction electrode **112***e* is located below the ring support surface **112***b*. The electrostatic chuck **112** attracts and holds the inner ring **113***a* on the ring support surface **112***b* by applying a voltage to the second attraction electrode **112***e*. In the examples in FIGS. **2** and **3**, the electrostatic chuck **112** includes a monopolar electrostatic chuck that attracts and holds the substrate W, and a bipolar electrostatic chuck that attracts and holds the inner ring **113***a*. A bipolar electrostatic chuck may be used instead of the monopolar electrostatic chuck, and a monopolar electrostatic chuck may be used instead of the bipolar electrostatic chuck.

[0047] The ring assembly **113** includes the inner ring **113***a* and the outer ring **113***b*. The inner ring **113***a* has an annular shape. The inner ring **113***a* is placed on the ring support surface **112***b* to surround the substrate W. The inner ring **113***a* improves uniformity of a plasma processing on the substrate W. The inner ring **113***a* is made of, for example, a conductive material such as silicon (Si) or silicon carbide (SiC). Further, the inner ring **113***a* may be made of an insulating material such as quartz. The outer ring **113***b* has an annular shape. The outer ring **113***b* is disposed on an outer peripheral portion of the inner ring **113***a*. The outer ring **113***b* protects an upper surface of the insulating member **115** from, for example, plasma. The outer ring **113***b* is made of, for example, an insulating material such as quartz. Further, the outer ring **113***b* may be made of a conductive material such as silicon or silicon carbide. In the illustrated example, an inner peripheral portion of the outer ring **113***b* is located inward of the outer peripheral portion of the inner ring **113***a*, the outer peripheral portion of the inner ring **113***a* is located outward of the inner peripheral portion of the outer ring **113***b*, and the inner ring **113***a* and the outer ring **113***b* partially overlap each other in top view. Accordingly, when a plurality of supporting pins **521** to be described later is raised and lowered, the outer ring **113***b* and the inner ring **113***a* are raised and lowered. The insulating member **115** is disposed to surround the lower electrode **111**. The insulating member **115** is fixed to a bottom portion of the plasma processing chamber **10**, to support the lower electrode **111**.

[0048] The upper electrode **12** along with an insulating member **13** constitutes the plasma processing chamber **10**. The upper electrode **12** supplies one or more types of processing gases from the gas supply **20** to the plasma processing space **10***s*. The upper electrode **12** includes a ceiling plate **121** and a support **122**. A lower surface of the ceiling plate **121** defines the plasma processing space **10***s*. A plurality of gas introduction ports **121***a* is formed in the ceiling plate **121**. Each of the plurality of gas introduction ports **121***a* penetrates through the ceiling plate **121** in a plate thickness direction (vertical direction) of the ceiling plate **121**. The support **122** detachably supports the ceiling plate **121**. A gas diffusion chamber **122***a* is provided inside the support **122**. A plurality of gas introduction ports **122***b* extends downward from the gas diffusion chamber **122***a*. The plurality of gas introduction ports **122***b* communicates with the plurality of gas introduction ports **121***a*, respectively. The support **122** is provided with a gas supply port **122***c*. The upper electrode **12** supplies one or more processing gases from the gas supply port **122***c* to the plasma processing space **10***s* via the gas diffusion chamber **122***a*, the plurality of gas introduction ports **122***b*, and the plurality of gas introduction ports **121***a*.

[0049] A sidewall of the plasma processing chamber **10** is provided with a loading/unloading port **10***p*. The substrate W is transferred between the plasma processing space **10***s* and an outside of the plasma processing chamber **10** via the loading/unloading port **10***p*. The loading/unloading port **10***p* is opened and closed by a gate valve G.

[0050] The gas supply **20** includes one or more gas sources **21** and one or more flow rate controllers **22**. The gas supply **20** supplies one or more types of processing gases from the respective gas sources **21** to the gas supply port **122***c* via the respective flow rate controllers **22**. The flow rate controller **22** may include, for example, a mass flow controller or a pressure-controlled flow rate controller. The gas supply **20** may include one or more flow rate modulation devices that modulate or pulse flow rates of one or more processing gases.

[0051] The RF power supply **30** includes two RF power sources (a first RF power source **31***a* and a second RF power source **31***b*) and two matchers (a first matcher **32***a* and a second matcher **32***b*). The first RF power source **31***a* supplies first RF power to the lower electrode **111** via the first matcher **32***a*. A frequency of the first RF power may be, for example, 13 MHz to 150 MHz. The second RF power source **31***b* supplies second RF power to the lower electrode **111** via the second matcher **32***b*. A frequency of the second RF power may be, for example, 400 kHz to 13.56 MHz. Instead of the second RF power source **31***b*, a DC power source may be used.

[0052] The exhaust system **40** may be connected to, for example, a gas exhaust port **10***e* provided at the bottom portion of the plasma processing chamber **10**. The exhaust system **40** may include a pressure adjusting valve and a vacuum pump. A pressure in the plasma processing space **10***s* is adjusted by the pressure adjusting valve. The vacuum pump may include a turbo molecular pump, a dry pump, or a combination thereof.

[0053] The lifter **50** includes a first lifter **51** and a second lifter **52**.

[0054] The first lifter **51** includes a plurality of supporting pins **511** and an actuator **512**. The plurality of supporting pins **511** is inserted into through-holes H1 formed in the lower electrode **111** and the electrostatic chuck **112** so as to be capable of protruding and retracting from an upper surface of the electrostatic chuck **112**. The plurality of supporting pins **511** protrudes from the upper surface of the electrostatic chuck **112** to support the substrate W with upper ends of the plurality of supporting pins **511**, which are in contact with a lower surface of the substrate W. The actuator **512** raises and lowers the plurality of supporting pins **511**. As the actuator **512**, for example, a motor such as a DC motor, a stepping motor, or a linear motor, an air driving mechanism such as an air cylinder, or a piezoelectric actuator may be used. For example, when transferring the substrate W between the transfer robot TR1 and the substrate support **11**, the first lifter **51** raises and lowers the plurality of supporting pins **511**.

[0055] The second lifter **52** includes the plurality of supporting pins **521** and an actuator **522**. The supporting pin **521** is a stepped supporting pin formed of cylindrical (solid rod-shaped) members. The supporting pin **521** has a lower pin **523** and an upper pin **524**. The upper pin **524** is installed on the lower pin **523**. An outer diameter of the lower pin **523** is larger than an outer diameter of the upper pin **524**. Accordingly, a stepped portion is formed by an upper end surface **523***a* of the lower pin **523**. The lower pin **523** and the upper pin **524** are, for example, integrally formed.

[0056] The supporting pin **521** is inserted into a through-hole H**11** formed in the lower electrode **111**, a through-hole H**12** formed in the insulating member **115**, and a through-hole H**13** formed in the outer ring **113***b* so as to be capable of protruding and retracting from the upper surface of the insulating member **115** and an upper surface of the outer ring **113***b*. Inner diameters of the through-holes H**11** and H**12** are slightly larger than the outer diameter of the lower pin **523**. An inner diameter of the through-hole H**13** is slightly larger than the outer diameter of the upper pin **524**, and is smaller than the outer diameter of the lower pin **523**.

[0057] The supporting pin **521** may be displaced between a standby position, a first support position, and a second support position.

[0058] The standby position is a position at which an upper end surface **524***a* of the upper pin **524** is below a lower surface of the inner ring **113***a*. When the supporting pin **521** is in the standby position, the inner ring **113***a* and the outer ring **113***b* are supported on the electrostatic chuck **112** and the insulating member **115**, respectively, without being lifted by the supporting pin **521**.

[0059] The first support position is a position above the standby position. The first support position is a position at which the upper end surface **524***a* of the upper pin **524** protrudes above the upper surface of the outer ring **113***b*, and the upper end surface **523***a* of the lower pin **523** is below a lower surface of the outer ring **113***b*. As the supporting pin **521** moves to the first support position, the upper end surface **524***a* of the upper pin **524** comes into contact with a recess formed in the lower surface of the inner ring **113***a* to support the inner ring **113***a*.

[0060] The second support position is a position above the first support position. The second support position is a position at which the upper end surface **523***a* of the lower pin **523** protrudes above the upper surface of the insulating member **115**. As the supporting pin **521** moves to the second support position, the upper end surface **524***a* of the upper pin **524** comes into contact with the recess to support the inner ring **113***a*, and the upper end surface **523***a* of the lower pin **523** comes into contact with the lower surface of the outer ring **113***b* to support the outer ring **113***b*.

[0061] The actuator **522** raises and lowers the plurality of supporting pins **521**. The actuator **522** may be configured similarly to the actuator **512**.

[0062] When transferring the inner ring **113***a* between the transfer robot TR**1** and the substrate support **11**, the second lifter **52** lifts the inner ring **113***a* by moving the plurality of supporting pins **521** to the first support position. When transferring the inner ring **113***a* and the outer ring **113***b* between the transfer robot TR**1** and the substrate support **11**, the second lifter **52** lifts the inner ring **113***a* and the outer ring **113***b* by moving the plurality of supporting pins **521** to the second support position. Alternatively, even when transferring the outer ring **113***b* between the transfer robot TR**1** and the substrate support **11** in a state in which the inner ring **113***a* does not exist, the second lifter **52** lifts the outer ring **113***b* by moving the plurality of supporting pins **521** to the second support position.

[0063] The controller **90** controls each part of the plasma processing apparatus **1**. The controller **90** includes, for example, a computer **91**. The computer **91** includes, for example, a CPU **911** which is a processor, a storage **912**, and a communication interface **913**. The CPU **911** is configured to perform various control operations, based on a program stored in the storage **912**. The storage **912** includes at least one memory type selected from a group consisting of auxiliary storage devices such as a RAM, a ROM, a hard disk drive (HDD), and a solid state drive (SSD). The communication interface **913** may communicate with the plasma processing apparatus **1** via a communication line such as a local area network (LAN). The controller **90** may be provided separately from the controller CU or may be included in the controller CU.

(Position Misalignment of Ring)

[0064] Next, a cause of a position misalignment which occurs in the ring **113** when taking out the ring **113** from the substrate support **11** is described with reference to FIG. **4**.

[0065] In the processing modules PM**1** to PM**7** (e.g., the plasma processing apparatus **1** in FIG. **2**), when processing the substrate W, the substrate W is electrostatically attracted and fixed by the electrostatic chuck **112**. Further, in the plasma processing apparatus **1**, at a periphery of the substrate W, the ring **113** (the inner ring **113***a*) is electrostatically attracted and fixed by the electrostatic chuck **112**.

[0066] In the plasma processing apparatus **1**, after a plasma processing, charges in the electrostatic chuck **112** are reduced by a charge removal processing (gas charge removal, plasma charge removal, or the like). Further, by applying a direct current voltage having a polarity different from that in the plasma processing to the electrostatic chuck **112** of the substrate support **11**, electrostatic forces of the substrate support surface **112***a* and the ring support surface **112***b* may be reduced.

[0067] Although the charge removal processing is performed, charges may slightly remain in the substrate support surface **112***a* and the ring support surface **112***b*. Therefore, even after the charge removal processing, attraction of the inner ring **113***a* to the ring support surface **112***b* may slightly occur (hereinafter, also referred to as residual attraction). When the inner ring **113***a* is spaced apart from the ring support surface **112***b* by raising the supporting pins **521** so as to exchange the inner ring **113***a*, the inner ring **113***a* vibrates due to the residual attraction of the inner ring **113***a*, and hence the inner ring **113***a* is placed on the supporting pins **521** in a state in which the inner ring **113***a* is misaligned from a normal position of the inner ring **113***a*. Accordingly, there occurs a phenomenon in which the inner ring **113***a* is placed in a position misaligned from an original holding position (the supporting pins **521**) at which the transfer robot TR**1** holds the inner ring **113***a*.

[0068] For example, as illustrated in the upper right drawing of FIG. **4**, the inner ring **113***a* may be misaligned in a horizontal direction (lateral direction). Accordingly, when transferring the inner ring **113***a* to the transfer robot TR**1**, the transfer robot TR**1** holds the inner ring **113***a* at a misaligned position. The misaligned inner ring **113***a* may cause a problem such as that, in transfer of the transfer robot TR**1**, the inner ring **113***a* is broken due to interference with other parts.

[0069] Further, for example, as illustrated in the lower right drawing of FIG. **4**, the position of the inner ring **113***a* is considerably misaligned due to the residual attraction, and therefore, the inner ring **113***a* may be obliquely inclined while getting out of some supporting pins **521** among the supporting pins **521**. In this case, the transfer robot TR**1** which slides in the horizontal direction above the substrate support **11** collides with the inner ring **113***a*, and therefore, the transfer robot TR**1** and the inner ring **113***a* may be broken. That is, in the substrate processing system PS, it is required for the transfer robot TR**1** to perform unloading or loading by suppressing the position misalignment of the inner ring **113***a*. However, in a conventional substrate processing system, when taking output an inner ring (edge ring) from a substrate support, whether a transfer robot held the inner ring at a normal position was not recognized.

[0070] When unloading the inner ring **113***a*, the controller CU of the substrate processing system PS according to this embodiment acquires an index related to a position misalignment amount of the inner ring **113***a*. Further, when the position of the inner ring **113***a* is misaligned, the controller CU performs correction for resolving the position misalignment. Next, a configuration of acquiring an index related to a position misalignment amount of the inner ring **113***a* is described with reference to FIGS. **5**A to **5**C.

[0071] FIG. **5**A is a view illustrating a first example of detecting an index related to a position misalignment amount of the inner ring **113***a*. The substrate processing system PS uses, as the index related to the position misalignment amount, a detection result of a position of the inner ring **113***a* detected by the position detection sensor S**1** installed on the upper fork FK**1** (or the position detection sensor S**2** installed on the lower fork FK**2**). The position detection sensors S**1** and S**2** may be installed on a lower surface of the upper fork FK**1** or the lower fork FK**2**, which faces the substrate support **11** (a surface opposite to a surface on which the substrate W is held). Further, a plurality of position detection sensors S**1** or S**2** may be installed at the upper fork FK**1** or the lower fork FK**2**. For example, the position detection sensor S**1** may be installed at each of two prong portions of the upper fork FK**1**. Further, an installation place of the position detection sensor S**1** or S**2** is not limited to the lower surface facing the substrate support **11** (the surface opposite to the surface on which the substrate W is held), and the position detection sensor S1 or S2 may be installed at a side surface of the upper fork FK1 or the lower fork FK2.

[0072] For example, an optical detector (e.g., a displacement sensor) capable of detecting a shape change of an object may be applied to the position detection sensors S1 and S2. The position detection sensor S1 or S2 detects an edge of the substrate support surface **112***a* by detecting a step difference between the substrate support surface **112***a* and the ring support surface **112***b* along with movement of the upper fork FK1 or the lower fork FK2 in the horizontal direction (left arrow of S1 in FIG. 5A). Further, an upper surface of the inner ring **113***a* includes an inner portion **113***a***1** lower than the substrate support surface **112***a* on an inner side in a radial direction and an outer portion **113***a***2** higher than the inner portion **113***a***1** on an outer at a diameter direction outer side of the inner portion **113***a***1**. The position detection sensor S1 or S2 detects an edge of the inner ring **113***a* by detecting a step difference between the inner portion **113***a***1** and the outer portion **113***a***2** along with movement of the upper fork FK1 or the lower fork FK2 in the horizontal direction (right arrow of S1 in FIG. 5A). The controller CU may obtain a gap amount between the substrate support surface **112***a* and the inner ring **113***a*, based on a horizontal position (X coordinate and Y coordinate) of the edge of the substrate support surface **112***a* and a horizontal position (X coordinate and Y coordinate) of the edge of the inner ring **113***a*. The gap amount is a length of a gap in the horizontal direction between an inner peripheral surface of the inner portion **113***a***1** of the inner ring **113***a* and a sidewall between the substrate support surface **112***a* and the ring support surface **112***b*. For example, the gap amount is calculated as a value decomposed into an amount in an X-axis direction and an amount in a Y-axis direction.

[0073] The gap amount is specified in advance according to an outer diameter of the substrate support surface **112***a* and an inner diameter of the inner ring **113***a*. When there exists a portion of an annular gap, at which an absolute value of the gap amount is larger than a specified value, a position misalignment amount of the inner ring **113***a* becomes large. As such, the controller CU may calculate the position misalignment amount of the inner ring **113***a* with a high degree of precision, based on the detection result of the position detection sensor S1 or S2 (an example of the index related to the position misalignment amount).

[0074] FIG. **5**B is a view illustrating a second example of detecting an index related to a position misalignment amount of the inner ring **113***a*. The substrate processing system PS may perform a gas leakage check between the ring support surface **112***b* and the inner ring **113***a* as the index related to the position misalignment amount.

[0075] That is, a gas supply port **61***a* is formed in the ring support surface **112***b* at a peripheral edge of the electrostatic chuck **112**. The gas supply port **61***a* supplies a gas into a gap between a rear surface of the inner ring **113***a* placed on the ring support surface **112***b* and the ring support surface **112***b*. The gas may be the same as a heat transfer gas supplied to a bottom surface of the inner ring **113***a* in a plasma processing. As an example of the gas, He gas may be applied. Further, in a gas flow path **61** that communicates with the gas supply port **61***a*, an end portion of the gas flow path **61**, which is opposite to the ring support surface **112***b*, is connected to a gas supply **66** via a pipe **62**. The gas supply **66** may include one or more gas sources **661** and one or more flow rate controllers **662**. In an embodiment, the gas supply **66** is configured to, for example, supply the gas from the gas source **661** to the gas supply port **61***a* via the flow rate controller **662**. Each flow rate controller **662** may include, for example, a mass flow controller or a pressure-controlled flow rate controller. The gas flow path **61** and the pipe **62** may function as at least a portion of a supply path that supplies the gas between the ring support surface **112***b* and the rear surface of the inner ring **113***a*.

[0076] Further, the end portion of the gas flow path **61**, which is opposite to the ring support surface **112***b*, is connected to an exhaust system **64** via the pipe **62**. Accordingly, a periphery of the ring support surface **112***b* of the electrostatic chuck **112** may be exhausted via the gas supply port **61***a*. That is, the gas supply port **61***a* may function as an exhaust hole that exhausts the periphery of the ring support surface **112***b*. Therefore, in the embodiment, the gas flow path **61** and the pipe **62** may function as at least a portion of an exhaust path that exhausts the gap between the ring support surface **112***b* and the rear surface of the inner ring **113***a*.

[0077] Further, as for the electrostatic chuck **112**, a pressure sensor **67** that measures a pressure of the gap between the inner ring **113***a* electrostatically attracted to the ring support surface **112***b* and the ring support surface **112***b* is provided. The pressure sensor **67** is, for example, installed in the pipe **62**. Further, a switching valve **65** that switches execution and stop of gas supply by the gas supply **66** may be installed in the pipe **62**. Similarly, a switching valve **63** that switches execution and stop of exhaust of the periphery of the ring support surface **112***b* by the exhaust system **64** may be installed in the pipe **62**.

[0078] In a gas leakage check, the controller CU of the substrate processing system PS operates, for example, a gas check mechanism **60** (which including the gas flow path **61**, the pipe **62**, the switching valve **63**, the exhaust system **64**, the switching valve **65**, the gas supply **66** and the pressure sensor **67**) and the electrostatic chuck **112** according to a sequence described below. First, the controller CU applies a voltage to the second attraction electrode **112***e* of the electrostatic chuck **112** in a state in which the substrate W does not exist in the plasma processing chamber **10**, the inner ring **113***a* is placed on the ring support surface **112***b*, and exhaust is performed via the gas flow path **61** by the exhaust system **64**. Accordingly, a direct current voltage (e.g., direct current voltages having different polarities in the case of bipolar second attraction electrodes **112***e*) is applied to the second attraction electrode **112***e* of the electrostatic chuck **112**.

[0079] Subsequently, the controller CU supplies a gas to the gas flow path **61** such that a pressure in the gas flow path **61** is maintained higher than a pressure in the plasma processing chamber **10**. Specifically, the controller CU closes the switching valve **63** and stops the exhaust via the gas flow path **61** by the exhaust system **64**. Meanwhile, the controller CU opens the switching valve **65** and supplies the gas by the gas supply **66**. The gas is supplied into a gap between the rear surface of the inner ring **113***a* and the ring support surface **112***b* via the pipe **62** and the gas flow path **61**. If a pressure in the gap becomes a target pressure (e.g., if a result measured by the pressure sensor **67** becomes the target pressure), the controller CU closes the switching valve **65** and stops the supply of the gas. The target pressure is, for example, the same as a pressure in the gap in the plasma processing.

[0080] After that, the controller CU measures a pressure in a flow path including the gas flow path **61** by the pressure sensor **67**. Specifically, the pressure sensor **67** measures a pressure in the pipe **62** after a predetermined time elapses from the stop of the supply of the gas. The measured pressure is substantially the same as the pressure in the gap between the ring support surface **112***b* and the inner ring **113***a*. Therefore, the controller CU may determine leakage of the gas from the gap (an index related to a position misalignment amount of the inner ring **113***a*), based on the pressure measured by the pressure sensor **67**. More specifically, the controller CU determines whether the measured pressure is less than a pressure threshold value as determination of gas leakage from the gap. The pressure threshold value is, for example, set to 90% to 98% of the target pressure, and information of the pressure threshold value is stored in advance in the storage **912**. If the measured pressure is less than the pressure threshold value, the controller CU determines that the gas is leaked from the gap between the ring support surface **112***b* and the inner ring **113***a*, i.e., that the position of the inner ring **113***a* is misaligned.

[0081] FIG. **5**C is a view illustrating a third example of detecting an index related to a position misalignment amount of the inner ring **113***a*. The controller CU acquires, as the index related to the position misalignment amount, image information of a camera CM installed on the upper fork FK**1** or a camera CM installed on the lower fork FK**2**. The camera CM is, for example, installed on the lower surface of the upper fork FK**1** or the lower fork FK**2**, which faces the substrate support **11**, instead of the position detection sensors S**1** and S**2**. Further, the camera CM may be installed on a side surface of the upper fork FK**1** or the lower fork FK**2**.

[0082] In the image information, for example, a gap between the edge of the substrate support surface **112***a* and the edge of the inner ring **113***a* is included as color or contrast information. Therefore, the controller CU acquires image information imaged by the camera CM (an example of the index related to the position misalignment amount), and performs an appropriate image processing on the image information, so that it is possible to calculate a gap amount (position misalignment amount) with a high degree of precision.

[0083] Back to FIG. **1**, the controller CU of the substrate processing system PS determines necessity of exchange of the inner ring **113***a*, based on a trigger such as an instruction by a user, a number of substrate processings, quality of the substrate W, a sensor value of each of the processing modules PM**1** to PM**7**, or error occurrence. When determining that the exchange is necessary, the controller CU collects the inner ring **113***a* of the processing module for an exchange target, and performs a processing of exchanging the inner ring **113***a* with an inner ring **113***a* for exchange, which is accommodated in the ring accommodation module RSM. The inner ring **113***a* for exchange may be a new product (not used) or may be one that is completely used but is not consumed so much.

[0084] Further, when taking out the inner ring **113***a* from the processing module for the exchange target, the controller CU acquires an index related to a position misalignment amount of the inner ring **113***a* as described above to recognize whether a misalignment has occurred in the inner ring. Accordingly, it is possible to for the substrate processing system PS to correct the misalignment of the inner ring **113***a*, and thus it is possible to avoid damage of a part and the like.

[Transfer Method]

[0085] FIG. **6** is a flowchart illustrating an operation sequence when taking out an edge ring ER. Subsequently, an operation until taking out the edge ring ER is described with reference to FIG. **6**. The edge ring ER corresponds to the inner ring **113***a* illustrated in FIGS. **3**, **4** and **5**A to **5**C. Hereinafter, a case where the inner ring **113***a* is transferred between the processing module PM**1** and the ring accommodation module RSM is described. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0086] The operation sequence includes steps S**1** to S**6**. The steps S**1** to S**6** are performed as the controller CU and/or the controller **90** controls each part of the substrate processing system PS. Although the controller CU is described separately from the controller **90**, the controller CU may have functions of the controller **90** to control all processings, and the controller **90** may have functions of the controller CU to control all processings.

[0087] In the step S**1**, the controller CU controls the plasma processing apparatus **1** constituting the processing module PM**1** to perform a plasma processing (substrate processing) on the substrate W in the plasma processing chamber **10**. In the plasma processing, the controller **90** of the plasma processing apparatus **1** attracts the substrate W and the edge ring ER by applying a direct current voltage to the first attraction electrode **112***d* and the second attraction electrode **112***e* of the electrostatic chuck **112**.

[0088] Then, after the plasma processing, the controller CU unloads the substrate W in the plasma processing apparatus **1** by the transfer robot TR**1**. At this time, the controller **90** stops the application of the direct current voltage to the first attraction electrode **112***d* (or applies another direct current voltage having a polarity different from a polarity of the direct current voltage) and enables the substrate W to be taken out. The controller **90** raises the substrate W by the first lifter **51** and transfers the substrate W to the transfer robot TR**1** which enters into the plasma processing chamber **10**. The transfer robot TR**1** unloads the substrate W from the plasma processing apparatus **1**, and transfers the substrate W to a next processing module or any one of the load lock modules LL**1** to LL**3**.

[0089] In the step S**2**, after unloading the substrate W, the controller CU controls the plasma processing apparatus **1** to operate in an idle mode. The idle mode is a mode after the substrate W is plasma-processed, and is a mode in which the plasma processing is not performed. In the idle mode, the controller CU determines necessity of exchange of the edge ring ER, and determines whether the exchange of the edge ring ER can be performed. Necessity of exchange of the ring **113** may be determined based on the above-described trigger. The controller CU determines whether the exchange can be performed, based on existence of the substrate W in the processing module PM**1** for the exchange target, a transfer schedule of the transfer robot TR**1**, or the like. Further, in the step S**2**, the controller **90** keeps attracting the inner ring **113***a* by applying the direct current voltage to the second attraction electrode **112***e*. The determination of the exchange of the edge ring ER is not limited to that in the idle mode, and may be performed at another timing such as a plasma processing.

[0090] When determining that the exchange of the inner ring **113***a* is performed, in the step S**3**, the controller CU controls the plasma processing apparatus **1** to perform a charge removal processing on the inner ring **113***a*. For example, when performing a gas charge removal processing in the charge removal processing, the plasma processing apparatus **1** controls the plasma processing chamber **10** at a predetermined pressure by supplying an inert gas such as N_2 gas into the plasma processing chamber **10** by the gas supply **20**, and exhausting a gas in the plasma processing chamber **10** by the exhaust system **40**. Further, the plasma processing apparatus **1** applies a direct current voltage having a polarity different from a polarity of the direct current voltage applied in the plasma processing to the second attraction electrode **112***e* of the electrostatic chuck **112** for a predetermined time, and then stops. Further, the plasma processing apparatus **1** stops the pressure control and ends the gas charge removal.

[0091] Further, in the steps S**4** to S**6**, the controller CU transfers the edge ring ER from the processing module PM**1** to the ring accommodation module RSM. First, a case where state identification and transfer of the edge ring ER is performed by the position detection sensor S**1** or S**2** of the transfer robot TR**1** illustrated in FIG. **5**A (hereinafter, also referred to as a transfer method according to a first example) is described.

[0092] In the step S**4**, in a case where the controller CU identifies a misalignment of the inner ring **113***a* when raising the inner ring **113***a* from the substrate support **11**, the controller CU performs a misalignment identification and correction processing of correcting the misalignment.

[0093] FIG. **7** is a flowchart illustrating a processing flow of the misalignment identification and correction processing. FIGS. **8**A to **8**F are views illustrating an operation of the misalignment identification and correction processing. In the misalignment identification and correction processing, first, the controller CU raises the supporting pins **521** of the plasma processing apparatus **1** and separates the edge ring ER from the ring support surface **112***b* of the substrate support **11** (step S**101** in FIG. **7**). When residual attraction exists between the ring support surface **112***b* and the edge ring ER, a misalignment may occur in the edge ring ER when raised as illustrated in FIG. **8**A.

[0094] The controller CU lowers the supporting pins **521** of the plasma processing apparatus **1** and temporarily places the edge ring ER on the ring support surface **112***b* of the substrate support **11** for the time being (step S**102** in FIG. **7**). When the misalignment of the edge ring ER occurs, the edge ring ER is disposed while being misaligned from the substrate support surface **112***a* as illustrated in FIG. **8**B.

[0095] Subsequently, the controller CU operates the transfer robot TR**1** and detects a position of the edge ring ER (an example of an index related to a position misalignment amount) by the position detection sensor S**1** or S**2** (step S**103** in FIG. **7**). For example, as illustrated in FIG. **8**C, the transfer robot TR**1** moves in the horizontal direction within the plasma processing space **10***s* to dispose the position detection sensor S**1** or S**2** at a position capable of detecting an edge of the substrate support surface **112***a* and an edge of the edge ring ER. Accordingly, the controller CU receives a detection result of the position detection sensor S**1** or S**2**, thereby obtaining gap amounts (position misalignment amounts) of the substrate support surface **112***a* and the edge ring ER.

[0096] Subsequently, the controller CU determines whether the position misalignment amount of the edge ring ER is a pre-stored threshold value or more by comparing the position misalignment amount of the edge ring ER with the threshold value (step S**104**). The threshold value may be set to an appropriate value according to a measurement of the substrate support surface **112***a* or the edge ring ER, and may be set to, for example, a value in a range of about 0.1 mm to about 0.4 mm. In this embodiment, the threshold value is set to 0.15 mm.

[0097] If the position misalignment amount is less than the threshold value (the step S**104**: "NO"), the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small). Therefore, the controller CU ends the misalignment identification and correction processing (the step S**4** in FIG. **6**), and proceeds to the step S**5**. Meanwhile, if the position misalignment amount is the threshold value or more (the step S**104**: "YES"), it may be determined that the edge ring ER is misaligned (or the position misalignment amount is large). In this case, the controller CU detects an error of the edge ring ER, and proceeds to step S**105** in FIG. **7**.

[0098] In the step S**105**, the controller CU raises the edge ring ER by the supporting pins **521** of the plasma processing apparatus **1**, and enters the transfer robot TR**1** into the plasma processing space **10***s* to transfer the edge ring ER. Accordingly, as illustrated in FIG. **8**D, the edge ring ER is held by the transfer robot TR**1** in a state in which the edge ring ER is misaligned.

[0099] After that, the controller CU performs a correction movement of correcting the misalignment of the edge ring ER by the transfer robot TR**1** that holds the edge ring ER (step S**106** in FIG. **7**). In a configuration that detects the misalignment of the edge ring ER by the position detection sensor S**1** or S**2**, in the step S**103**, the position misalignment amount of the edge ring ER has already been acquired. Therefore, as illustrated in FIG. **8**E, the controller CU sets a movement amount in the correction movement of the transfer robot TR**1** based on the acquired position misalignment amount, and horizontally moves the transfer robot TR**1** by the movement amount. In the example of FIG. **8**E, the transfer robot TR**1** is horizontally moved in a right direction by the movement amount. As a result, the edge ring ER is returned to a normal position above the substrate support **11**.

[0100] The controller CU receives the edge ring ER from the transfer robot TR**1** by the supporting pins **521**, and places the edge ring ER on the ring support surface **112***b* again by lowering the supporting pins **521** after withdrawal of the transfer robot TR**1** (step S**107** in FIG. **7**). Accordingly, as illustrated in FIG. **8**F, the edge ring ER is supported on the ring support surface **112***b* in a state in which the position misalignment is resolved. When the step S**107** is ended, the misalignment identification and correction processing (step S4) is ended. Further, after receiving the edge ring ER by the supporting pins **521**, the controller CU may perform an unloading processing of unloading the edge ring ER from the processing module PM**1** without lowering the supporting pins **521**.

[0101] Back to FIG. **6**, in the step S**5**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1**. Specifically, the controller CU raises the edge ring ER by the supporting pins **521** of the plasma processing apparatus **1** and receives the edge ring ER via the transfer robot TR**1**. After placing the edge ring ER (after the step S**107**), electrostatic attraction by the electrostatic chuck **112** is not performed. Therefore, when raising the edge ring ER by the supporting pins **521**, the edge ring ER may be disposed without being misaligned from a holding position of the transfer robot TR**1**. The transfer robot TR**1** receives the edge ring ER in which no misalignment occurs, and unloads the edge ring ER from the processing module PM**1** after the lowering of the supporting pins **521**. At this time, the transfer robot TR**1** may smoothly unload the edge ring ER without interfering with other parts.

[0102] In the step S**6**, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by transferring the edge ring ER by the transfer robot TR**1**. Since the transfer robot TR**1** holds the edge ring ER in which no misalignment occurs, it is possible to stably transfer the edge ring ER into the ring accommodation module RSM without interfering with other parts of the ring accommodation module RSM or the like.

[0103] As described above, in the substrate processing system PS and the transfer method, a misalignment of the edge ring ER is identified when unloading the edge ring ER from the processing module PM**1**, and a correction is performed when the misalignment occurs. Therefore, it is possible to transfer the edge ring ER without interfering with other parts. Accordingly, it is possible to prevent the edge ring ER from dropping off the transfer robot TR**1**, and to avoid damage to the edge ring ER or other parts, and the like. Thus, it is possible to for the substrate processing system PS to reduce an opportunity for stopping the substrate processing system PS for performing a maintenance.

[0104] Further, the substrate processing system PS and the transfer method are not limited to the embodiment, and may have various modifications. For example, the transfer method illustrated in FIG. **6** has described a case where only the edge ring ER is exchanged, but may be applied even in a case where a cover ring CR (the outer ring **113***b*) is exchanged. By performing a misalignment identification and correction processing even when unloading the cover ring CR, it is possible to stably transfer the cover ring CR.

[0105] The substrate support **11** of each of the processing modules PM**1** to PM**7** is not limited to a configuration that electrostatically attracts the substrate W and the ring **113** by the electrostatic chuck **112**. For example, the substrate support **11** may employ a mechanism that provides an attractive force to the substrate W and the ring **113**, a mechanism that mechanically hangs and fixes the ring **113**, or the like. In this case, since the ring **113** is adhered to the ring support surface **112***b* or other parts, the ring **113** may be misaligned when the supporting pins **521** are raised. Thus, it is possible for the substrate processing system PS to transfer the ring **113** while suppressing the misalignment of the ring **113** by performing the transfer method according to the embodiment.

[0106] In addition, in the misalignment identification and correction processing, after placing the edge ring ER on the ring support surface **112***b* in the step S**102**, the index related to the position misalignment amount of the edge ring ER is acquired by the position detection sensor S**1** or S**2** of the transfer robot TR**1**. However, in the misalignment identification and correction processing, the index related to the position misalignment amount of the edge ring ER may be acquired by the position detection sensor S**1** or S**2** of the transfer robot TR**1** in a state in which the edge ring ER is supported (raised) by the supporting pins **521**.

[0107] For example, as illustrated in FIG. **9**, in step S**111**, the controller CU raises the edge ring ER by the supporting pins **521**, and stops the supporting pins **521** at a timing at which the edge ring ER is separated from the ring support surface **112***b*. Further, the controller CU acquires a position misalignment amount by the position detection sensors S**1** and S**2** in a state in which the supporting pins **521** are stopped. Alternatively, the controller CU may acquire a position misalignment amount by the position detection sensors S**1** and S**2** at an arbitrary pin height which is higher than the position at which the edge ring ER is separated from the ring support surface **112***b* and at which the upper surface of the edge ring ER is below the position detection sensors S**1** and S**2** of the transfer robot TR**1**. In step S**112**, the controller CU detects a position misalignment amount of the edge ring ER by the position detection sensor S**1** or S**2**. In step S**113**, the controller CU determines whether the position misalignment amount of the edge ring ER is a threshold value or more. Further, if the position misalignment amount is the threshold value or more (the step S**113**: "YES"), the controller CU raises the edge ring ER to a transfer height by the supporting pins **521** and enters the transfer robot TR**1** into the plasma processing space **10***s* to transfer the edge ring ER (step S**114**). In step S**115**, the controller CU performs a correction movement of correcting a misalignment of the edge ring ER by the transfer robot TR**1** that holds the edge ring ER. Further, in step S**116**, the controller CU receives the edge ring ER from the transfer robot TR**1** by the supporting pins **521**, and places the edge ring ER on the ring support surface **112***b* again by lowering the supporting pins **521** after withdrawal of the transfer robot TR**1**. Accordingly, operations of lowering the edge ring ER in the step S**102** in FIG. **7** and raising the edge ring ER in the step S**105** in FIG. **7** may be omitted, thus more efficiently performing a processing. Further, in the step S**115**, the controller CU may move and correct the transfer robot TR**1** in consideration of the position misalignment amount of the edge ring ER, and then receive the edge ring ER and unload the edge ring ER as it is. Accordingly, the step S**116** may be omitted.

[0108] In addition, in the misalignment identification and correction processing, the misalignment of the edge ring ER has been resolved by receiving the edge ring ER by the transfer robot TR**1**, and then performing the correction movement. However, in the misalignment identification and correction processing, before the transfer robot TR**1** receives the edge ring ER held by the supporting pins **521** in the step S**105** in FIG. **7**, the transfer robot TR**1** may be corrected and moved in advance by the position misalignment amount, and then the edge ring ER may be transferred to the transfer robot TR**1**. Accordingly, for example, the step S**106** or S**107** in FIG. **7** may be omitted, thus much more efficiently performing a processing.

[0109] In addition, in the misalignment identification and correction processing (the step S**4**), the substrate processing system PS may perform a cleaning processing in the plasma processing chamber **10** by stopping at a predetermined height while raising the edge ring ER by the supporting pins **521**, may perform the cleaning processing in the plasma processing chamber **10** after raising the supporting pins **521** to the transfer height, or may perform the cleaning processing in the plasma processing chamber **10** while raising the edge ring ER. Alternatively, in the unloading processing from any one of the processing modules PM1 to PM7 (the step S5), the substrate processing system PS may perform the cleaning processing in the plasma processing chamber **10** by stopping at a predetermined height while raising the edge ring ER by the supporting pins **521**, may perform the cleaning processing in the plasma processing chamber **10** after raising the supporting pins **521** to the transfer height, or may perform the cleaning processing in the plasma processing chamber **10** while raising the edge ring ER. The cleaning processing may use, for example, a method such as wafer-less dry cleaning (WLDC) or a wafer with dry cleaning (WWDC) performed by placing, on the substrate support **11**, a dummy wafer that has a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface **112***a*. In the case of the WWDC, the dummy wafer having the same diameter as the substrate support surface **112***a* is placed on the substrate support **11** until before the steps S**4** and S**5**.

[0110] In addition, from after the substrate processing in the step S**1** to the charge removal processing in the step S**3**, the substrate processing system PS may perform wafer-less dry cleaning, or may perform dry cleaning by placing a dummy wafer having the same diameter as the substrate W or a dummy wafer having a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface **112***a*. Further, before the ring for exchange is loaded after the step S**5**, the substrate processing system PS may perform wafer-less dry cleaning, or may perform dry cleaning by placing a dummy wafer having the same diameter as the substrate W or a dummy wafer having a diameter smaller than the diameter of the substrate W and equal to the diameter of the substrate support surface **112***a*.

[0111] The cleaning processing is performed in the misalignment identification and correction processing or the unloading processing, so that it is possible to clean the edge ring ER before unloading the edge ring ER to the vacuum transfer chamber, thereby suppressing the vacuum transfer chamber from being contaminated by a deposit attached to the edge ring ER. In addition, the cleaning processing is performed at a timing at which the edge ring ER is raised, so that it is possible to remove a deposit attached to the ring support surface **112***b*. In addition, the substrate processing system PS performs the cleansing processing on the edge ring ER during an unloading operation, so that it is possible to improve a throughput of the entire processing as compared with a case where the cleaning processing is separately performed. In addition, the deposit attached to the ring support surface is removed by performing the cleaning processing after the edge ring is unloaded and before the edge ring for exchange is loaded, so that it is possible to suppress an attraction failure of the edge ring for exchange.

[0112] FIG. **10** is a flowchart illustrating a transfer method according to a second example. The transfer method according to the second example performs state identification and transfer of the edge ring ER by the position detection sensor S**1** or S**2** of the transfer robot TR**1**. Further, in description of the transfer method according to the second example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM**1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0113] When performing the steps to the step S**3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps S**201** to S**210** as the transfer method according to the second example. The steps S**201** to S**210** are performed as the controller CU controls each part of the substrate processing system PS.

[0114] The steps S**201** to S**204** may be the same as the steps S**101** to S**104**. However, if a position misalignment amount of the edge ring ER is a threshold value or more in the step S**204** (the step S**204**: "YES"), the controller CU may proceed to the step S**205**. If the position misalignment amount of the edge ring ER is less than the threshold value, the controller CU may proceeds to the step S**208** (the step S**204**: "NO").

[0115] In the step S**205**, the controller CU raises the edge ring ER by the supporting pins **521**, enters the transfer robot TR**1** into the plasma processing space **10***s*, and then lowers the supporting pins **521** to transfer the edge ring ER to the transfer robot TR**1**. Accordingly, the edge ring ER is held by the transfer robot TR**1** in a state in which the edge ring ER is misaligned.

[0116] After that, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1** (the step S**206**). By performing the unloading processing after the step S**205**, the transfer robot TR**1** unloads the edge ring ER from the processing module PM**1** while holding the edge ring ER in the state in which the edge ring ER is misaligned.

[0117] Further, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by transferring the edge ring ER by the transfer robot TR**1** (the step S**207**). In the loading processing, when loading the edge ring ER into an accommodation module (e.g., a cassette which is not illustrated) of the ring accommodation module RSM, the controller CU sets a movement amount of the transfer robot TR**1**, based on the position misalignment amount acquired in the step S**203**, and corrects and moves the transfer robot TR**1**. Accordingly, the edge ring ER is returned to the normal position by the transfer robot TR**1** without interfering with the ring accommodation module RSM. Thus, the controller CU may smoothly accommodate the edge ring ER without interfering with other parts.

[0118] Meanwhile, if the position misalignment amount is less than the threshold value, the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small), and hence the controller CU performs ordinary transfer (of not performing movement and correction) in the steps S**208** to S**210**.

[0119] Specifically, the controller CU raises the edge ring ER by the supporting pins **521** of the plasma processing apparatus **1** and enters the transfer robot TR**1** into the plasma processing space **10***s* to transfer the edge ring ER (the step S**208**). Further, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1** (the step S**209**). Further, the controller CU transfers the edge ring ER by the transfer robot TR**1**, and loads the edge ring ER into the ring accommodation module RSM (the step S**210**). When loading the edge ring ER into the ring accommodation module RSM, the transfer robot TR**1** may accommodate the edge ring ER in the accommodation module without moving and correcting the edge ring ER. That is, the unloading processings in the steps S**206** and S**209** correspond to the unloading processing in the step S**5** in FIG. **6**, and the loading processings in the steps S**207** and S**210** correspond to the loading processing in the step S**6** in FIG. **6**.

[0120] As such, in a case where the edge ring ER is misaligned, the transfer method according to the second example performs movement and correction when the edge ring ER is loaded into the ring accommodation module RSM, thus suppressing interference between the edge ring ER and other parts. Further, the transfer method according to the second example may also have various modifications. For example, in the misalignment identification and correction processing, an index related to a position misalignment amount of the edge ring ER may be acquired by the position detection sensor S**1** or S**2** of the transfer robot TR**1** in a state in which the edge ring ER is supported by the supporting pins **521**. That is, in the step S**201**, the controller CU may stop the supporting pins **521** at a position at which the edge ring ER is separated from the ring support surface **112***b* and then acquire a position misalignment amount by the position detection sensor S**1** or S**2**, or may acquire a position misalignment amount by the position detection sensors S**1** and S**2**, at an arbitrary pin height which is higher than the position at which the edge ring ER is separated from the ring support surface **112***b* and at which the upper surface of the edge ring ER is below the position detection sensors S**1** and S**2** (see FIG. **9**).

[0121] Further, the controller CU may select implementation of the transfer method according to the first example or implementation of the transfer method according to the second example based on the position misalignment amount of the edge ring ER. For example, the controller CU has a threshold value for selection of a transfer method, and may determine the transfer method according to the first example if the position misalignment amount is equal to or larger than the threshold value for selection, or may determine the transfer method according to the second example if the position misalignment amount is less than the threshold value for selection. The threshold value for selection is a value larger than a threshold value for determining movement and correction. Accordingly, when the position misalignment amount of the edge ring ER is large, a misalignment of the edge ring ER is corrected in the processing module PM**1**, so that more stable transfer is possible. Meanwhile, when the position misalignment amount of the edge ring ER is small even though movement and correction is necessary, a misalignment of the edge ring ER is corrected when the edge ring ER is transferred to the ring accommodation module RSM, so that it is possible to improve transfer efficiency.

[0122] The substrate processing system PS is not limited to a configuration of loading the ring **113** (the edge ring ER) in any one of the processing modules PM**1** to PM**7** directly into the ring accommodation module RSM by the transfer robot TR**1**. The substrate processing system PS may transfer an upper ring **222** of a ring **220** to a standby transfer module LM, for example, when a plasma processing apparatus **1**A shown in FIG. **11** is applied. FIG. **11** is a schematic cross-sectional view illustrating another example of the plasma processing apparatus.

[0123] The plasma processing apparatus **1**A shown in FIG. **11** is different from the plasma processing apparatus **1** in that the plasma processing apparatus **1**A includes a substrate support **16** instead of the substrate support **11** and includes a ring assembly **220** (ring **220**) instead of the ring assembly **113**. Other parts may be the same as the plasma processing apparatus **1**. Hereinafter, differences from the plasma processing apparatus **1** are mainly described.

[0124] The plasma processing apparatus **1**A includes the substrate support **16**. The substrate support **16** is installed inside the plasma processing chamber **10**. The substrate support **16** supports a substrate W. The substrate support **16** is supported by a support **17**. The support **17** extends upward from a bottom portion of the plasma processing chamber **10**. The support **17** has a cylindrical shape. The support **17** is made of an insulating material such as quartz.

[0125] The substrate support **16** has a first region **161** and a second region **162**. The first region **161** supports the substrate W. The first region **161** is a substantially circular region in a plan view. The first region **161** may include a base **18** and an electrostatic chuck **19**. The first region **161** may be configured by a portion of the base **18** and a portion of the electrostatic chuck **19**. The base **18** and the electrostatic chuck **19** are installed inside the plasma processing chamber **10**. The base **18** is made of a conductive material such as aluminum. The base **18** has a substantially disk shape. The base **18** constitutes a lower electrode.

[0126] The substrate support **16** includes a main body **2** and the ring assembly **220**. The main body **2** includes the base **18** and the electrostatic chuck **19**. The main body **2** has a substrate support region **2***a* for supporting the substrate W, an annular region **2***b* for supporting the ring assembly **220**, and a sidewall **2***c* that extends vertically between the substrate support region **2***a* and the annular region **2***b*. The annular region **2***b* surrounds the substrate support region **2***a*. The annular region **2***b* is located below the substrate support region **2***a*. Therefore, an upper end of the sidewall **2***c* is connected to the substrate support region **2***a*, and a lower end of the sidewall **2***c* is connected to the annular region **2***b*.

[0127] A flow path **18***f* is formed in the base **18**. The flow path **18***f* is a flow path through which a heat exchange medium flows. As the heat exchange medium, a liquid refrigerant or a refrigerant (e.g., chlorofluorocarbon) that cools the base **18** by vaporization of the liquid refrigerant is used. A supply device of the heat exchange medium (e.g., a chiller unit) is connected to the flow path **18***f*. The supply device is installed outside the plasma processing chamber **10**. The heat exchange medium is supplied to the flow path **18***f* from the supply device. The heat exchange medium supplied to the flow path **18***f* is returned to the supply device.

[0128] The electrostatic chuck **19** is installed on the base **18**. The substrate W is placed on the first region **161** and on the electrostatic chuck **19** when processed in the plasma processing chamber **10**.

[0129] The second region **162** extends radially outward from the first region **161** to surround the first region **161**. The second region **162** is a substantially annular region in a plan view. The ring assembly **220** is placed on the second region **162**. The second region **162** may include the base **18**. The second region **162** may include the electrostatic chuck **19**. The second region **162** may be composed of another portion of the base **18** and another portion of the electrostatic chuck **19**. The substrate W is placed in a region surrounded by the ring assembly **220** and on the electrostatic chuck **19**. Details of the ring assembly **220** will be described later.

[0130] A through-hole **162***h* is formed in the second region **162**. The main body **2** has the through-hole **162***h* formed between the annular region **2***b* and a lower surface **2***d* of the main body **2**. The through-hole **162***h* is formed in the second region **162** to extend along a vertical direction. A plurality of through-holes **162***h* are formed in the second region **162**. A number of the through-holes **162***h* may be equal to a number of lift pins **72** of a lift mechanism **70** which will be described later. Each through-hole **162***h* is disposed parallel with a corresponding lift pin **72** on a straight line.

[0131] The electrostatic chuck **19** includes a main body **19***m* and an electrode **19***e*. The main body **19***m* is made of a dielectric such as aluminum oxide or aluminum nitride. The main body **19***m* has a substantially disk shape. The electrode **19***e* is installed in the main body **19***m*. The electrode **19***e* has a film shape. A direct current power source is electrically connected to the electrode **19***e* via a switch. When a voltage from the direct current power source is applied to the electrode **19***e*, an electrostatic attraction force is generated between the electrostatic chuck **19** and the substrate W. By the generated electrostatic attraction force, the substrate W is attracted to the electrostatic chuck **19** and held by the electrostatic chuck **19**.

[0132] The plasma processing apparatus **1**A further includes an outer peripheral member **27**. The outer peripheral member **27** extends circumferentially and radially outward from the substrate support **16** to surround the substrate support **16**. The outer peripheral member **27** may extend circumferentially and radially outward from the support **17** to surround the support **17**. The outer peripheral member **27** may be composed of one or more parts. The outer peripheral member **27** may be formed of an insulating material such as quartz.

[0133] The ring assembly **220** and the substrate support **16** will now be described in more detail. The ring assembly **220** includes a lower ring **221** and an upper ring **222**.

[0134] Each of the lower ring **221** and the upper ring **222** has an annular shape. Each of the lower ring **221** and the upper ring **222** is formed of a material appropriately selected according to a plasma processing performed in the plasma processing apparatus **1**A. Each of the lower ring **221** and the upper ring **222** is formed of, for example, silicon or silicon carbide.

[0135] The lower ring **221** is disposed on the annular region **2***b*. The lower ring **221** may be placed on the second region **162** and on the electrostatic chuck **19**. The lower ring **221** may be placed on a part other than the electrostatic chuck **19** in the second region **162**.

[0136] A lower surface of the upper ring **222** is substantially flat. The lower surface of the upper ring **222** includes a tapered surface and defines a recess. The lower surface of the upper ring **222** defines a plurality of recesses. A number of tapered surfaces of the upper ring **222** and a number of recesses may be the same as the number of lift pins **72** of the lift mechanism **70**. Each recess has a size in which a tip end of a second rod portion **722** of a corresponding lift pin **72** is fitted into the recess. The upper ring **222** is disposed on the lower ring **221** such that each recess is aligned with corresponding lift pins **72** and corresponding through-holes **221***h* on a straight line.

[0137] The upper ring **222** is accommodated in a recess of the lower ring **221**. The lower ring **221** and the upper ring **222** are configured such that when disposed on the annular region **2***b*, an upper surface of an outer portion of the lower ring **221** and an upper surface of the upper ring **222** are substantially at the same height as an upper surface of the substrate W on the substrate support region **2***a*. The upper ring **222** includes an inner peripheral surface **222***a* that faces an end surface of the substrate W on the substrate support region **2***a* when the lower ring **221** and the upper ring **222** are disposed on the annular region **2***b*.

[0138] The substrate support **16** includes the lift mechanism **70**. The lift mechanism **70** includes the lift pins **72**, and is configured to raise and lower the lower ring **221** and the upper ring **222**. The lift mechanism **70** includes a plurality of lift pins **72**. The number of lift pins **72** may be any number as long as the lift pins **72** are capable of supporting, raising and lowering the ring assembly **220**. The number of lift pins **72** may be, for example three.

[0139] Each lift pin **72** may be formed of an insulating material. Each lift pin **72** may be formed of, for example, sapphire, alumina, quartz, silicon nitride, aluminum nitride, or resin. Each lift pin **72** includes a first rod portion **721** and the second rod portion **722**. The first rod portion **721** extends in the vertical direction. The first rod portion **721** includes a first upper end surface **721***t*. The first upper end surface **721***t* may come in contact with a lower surface of the lower ring **221**.

[0140] The second rod portion **722** extends vertically upward from the first rod portion **721**. The second rod portion **722** is narrowed with respect to the first rod portion **721** to expose the first upper end surface **721***t*. Each of the first rod portion **721** and the second rod portion **722** has a columnar shape. A diameter of the first rod portion **721** is larger than a diameter of the second rod portion **722**. The second rod portion **722** is capable of moving up and down through the through-hole **221***h*. A length of the second rod portion **722** in the vertical direction is longer than a vertical thickness of a region of the lower ring **221** on which the upper ring **222** is placed.

[0141] The second rod portion **722** has a second upper end surface **722***t*. The second upper end surface **722***t* may come in contact with the upper ring **222**. The tip end of the second rod portion **722** including the second upper end surface **722***t* may be formed in a tapered shape to be fitted into a corresponding recess in the upper ring **222**.

[0142] The second rod portion **722** may include a first portion **722***a* and a second portion **722***b*. The first portion **722***a* has a columnar shape and extends upward from the first rod portion **721**. The second portion **722***b* has a columnar shape and extends upward from the first portion **722***a*. The second portion **722***b* includes the second upper end surface **722***t*. A width of the first portion **722***a* is larger than a width of the second portion **722***b*.

[0143] The first rod portion **721**, the first portion **722***a*, and the second portion **722***b* may have a columnar shape. A diameter of the first rod portion **721** is larger than a diameter of the first portion **722***a*, and the diameter of the first portion **722***a* is larger than a diameter of the second portion **722***b*.

[0144] The second rod portion **722** may include a third portion **722***c*. The third portion **722***c* extends between the first portion **722***a* and the second portion **722***b*. The third portion **722***c* has a tapered surface.

[0145] The lift mechanism **70** includes one or more drives **74**. The one or more drives **74** are configured to raise and lower the plurality of lift pins **72**. Each of the one or more drives **74** may include, for example, a motor.

[0146] For example, the upper ring **222** is stored in a ring accommodation container CS**2** loaded into the load port LP**4** of the atmospheric transfer module LM. The ring accommodation container CS**2** corresponds to a ring accommodation module. The controller CU unloads the upper ring **222** of any one of the processing modules PM**1** to PM**7** to the vacuum transfer module TM, and transfers the upper ring **222** to any one of the load lock modules LL**1** to LL**3**. Further, when unloading the upper ring **222** from any one of the load lock modules LL**1** to LL**3** using the transfer robot TR**2** of the atmospheric transfer module LM, the controller CU loads and accommodate the upper ring **222** in the ring accommodation container CS**2**.

[0147] Correction and movement when a position misalignment occurs in the upper ring **222** may be performed in any one of the processing modules PM**1** to PM**7**, or may be performed when the transfer robot TR**1** loads and places the upper ring **222** in any one of the load lock modules LL**1** to LL**3**. Alternatively, the correction and movement may be performed when the transfer robot TR**2** of the atmospheric transfer module LM loads the upper ring **222** into the ring accommodation container CS**2**. Further, in the transfer method, the same processing as above may be performed even when transferring the lower ring **221** in addition to the upper ring **222**. In addition, for example, the processing modules PM**1** to PM**7** of the substrate processing system PS are not limited to the ring assemblies **113** and **220** including a combination of a plurality of members, and may be configured to use a ring including one member. In this case, the transfer methods according to the first and second examples may be applied, and transfer methods according to third to fifth examples, which will be described later, may be applied.

[0148] Next, a transfer method when performing state identification and transfer of the edge ring ER by the leakage check illustrated FIG. **5**B is described. FIG. **12** is a flowchart illustrating a processing flow in a transfer method according to a third example. Further, in description of the transfer method according to the third example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM**1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0149] When performing the steps to the step S**3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps S**301** to S**308** as the transfer method according to the third example. The steps S**301** to S**308** are performed as the controller CU controls each part of the substrate processing system PS.

[0150] The steps S**301** and S**302** may be the same as the steps S**101** and S**102**. In addition, in order to perform a leakage check after the step S**302** (before the step S**303**), the controller CU performs a processing of electrostatically attracting the edge ring ER to the ring support surface **112***b*.

[0151] In the step S**303**, in order to acquire an index related to a position misalignment amount of the edge ring ER, the controller CU performs a leakage check by the plasma processing apparatus **1**. As illustrated in FIG. **5**B, in the leakage check, the controller **90** of the plasma processing apparatus **1** supplies a gas to a gap between the ring support surface **112***b* to which the edge ring ER is electrostatically attracted and the edge ring ER, and stops the supply of the gas when a pressure in the gap reaches a target pressure. Further, the controller **90** measures a pressure in a flow path after a predetermined time elapses from the stop of the supply by the pressure sensor **67**.

[0152] The controller CU acquires, as the index related to the position misalignment amount of the edge ring ER, a measurement result of the pressure sensor **67** after the predetermined time elapses, and determines a position misalignment of the edge ring ER (the step S**304** in FIG. **12**). For example, if the pressure is a pre-stored pressure threshold value or more, the controller CU determines that the edge ring ER is not misaligned (normal). Meanwhile, if the pressure less than the pressure threshold value, the controller CU determines that the edge ring ER is misaligned (abnormal). Further, if the edge ring ER is not misaligned (the step S**304**: "YES"), the controller CU proceeds to the step S**305**. If the edge ring ER is misaligned (the step S**304**: "NO"), the controller CU proceeds to the step S**308**.

[0153] In the step S**305**, the controller CU ends the leakage check and the misalignment identification and correction processing. Further, in the step S**306**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1** by the transfer robot TR**1**. The step S**306** corresponds to the step S**5** in FIG. **6**. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER before the step S**306** (or the step S**308** to be described later). Further, in the step S**307**, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR**1**. The step S**307** corresponds to the step S**6** in FIG. **6**.

[0154] Meanwhile, when a misalignment occurs in the edge ring ER, in the step S**308**, the controller CU performs a misalignment resolution operation for resolving the misalignment of the edge ring ER. In the misalignment resolution operation, for example, as illustrated in FIGS. **13**A to **13**C, a recess **114** formed in the edge ring ER and a configuration of each supporting pin **521** may be used.

[0155] Specifically, in a lower surface of the edge ring ER, the recess **114** of the edge ring ER is provided as many as the supporting pins **521**. As illustrated in FIG. **13**A, each recess **114** includes a flat bottom portion **114***a* and a tapered portion **114***b* having an inner diameter widened toward an opening portion while surrounding a periphery of the bottom portion **114***a*. The bottom portion **114***a* is formed wider than the outer diameter of the upper pin **524** of the supporting pin **521**, and have, for example, about 2 mm. The tapered portion **114***b* is smoothly connected to the bottom portion **114***a* and the lower surface of the edge ring ER through rounded edge portions.

[0156] When a position misalignment occurs on the edge ring ER having the recess **114**, for example, the upper end surface **524***a* of the supporting pin **521** and the tapered portion **114***b* face each other. In this case, the controller CU performs, as the misalignment resolution operation, an operation of raising the supporting pins **521** by the plasma processing apparatus **1**.

[0157] That is, as illustrated in FIG. **13**B, when raising the supporting pin **521**, the upper end surface **524***a* of the supporting pin **521** comes in contact with the tapered portion **114***b* of the edge ring ER. Therefore, the tapered portion **114***b* slides along a slope of the tapered portion **114***b* with respect to the supporting pin **521**. As a result, as illustrated in FIG. **13**C, the edge ring ER moves in a horizontal direction to guide the supporting pin **521** to a substantially central portion of the bottom portion **114***a*. By the movement in the horizontal direction, the misalignment of the edge ring ER is resolved.

[0158] Back to FIG. **12**, when performing the misalignment resolution operation in the step S**308**, the controller CU returns to the step S**303** to perform the leakage check again. This is for the purpose of identifying whether the misalignment of the edge ring ER has been resolved by the misalignment resolution operation. If the measured pressure is less than the pressure threshold value, the misalignment of the edge ring ER is not resolved, and hence the controller CU repeats the misalignment resolution operation (the step S**308**). Meanwhile, if the measured pressure is the pressure threshold value or more, the controller CU proceeds to the step S**305** to end the misalignment identification and correction processing.

[0159] As described above, in the substrate processing system PS and the transfer method, it is possible to identify the misalignment of the edge ring ER by performing the leakage check. Further, in the substrate processing system PS and the transfer method, it is possible to correct the misalignment of the edge ring ER by using the recess **114** of the edge ring ER and the shape of the supporting pin **521**. As a result, the transfer robot TR**1** may transfer the edge ring ER without interfering with other parts.

[0160] FIG. **14** is a flowchart illustrating a transfer method according to a fourth example. The transfer method according to the fourth example performs state identification and transfer of the edge ring ER by performing a leakage check in the processing module PM**1**. Further, in description of the transfer method according to the fourth example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM**1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0161] When performing the steps to the step S**3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps S**401** to S**410** as the transfer method according to the fourth example. The steps S**401** to S**410** are performed as the controller CU controls each part of the substrate processing system PS.

[0162] The steps S**401** to S**404** may be the same as the steps S**301** to S**304**. In addition, in order to perform a leakage check after the step S**402** (before the step S**403**), the controller CU performs a processing of electrostatically attracting the edge ring ER to the ring support surface **112***b*. Further, if the pressure of the pressure sensor **67** is less than the pressure threshold value in the step S**404** (the step S**404**: "NO"), the controller CU proceeds to the step S**405**. If the pressure of the pressure sensor **67** is the pressure threshold value or more (the step S**404**: "YES"), the controller CU omits the steps S**405** to S**408** and proceeds to the step S**409**.

[0163] In the step S**405**, the controller CU operates the transfer robot TR**1** and detects a position (an index related to a position misalignment amount) of the edge ring ER by the position detection sensor S**1** or S**2**. Accordingly, the controller CU may obtain a gap amount (a position misalignment amount) between the substrate support surface **112***a* and the edge ring ER by receiving a detection result of the position detection sensor S**1** or S**2**.

[0164] Further, controller CU may perform correction and movement illustrated in of FIGS. **8**D to **8**F by utilizing the position misalignment amount. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER from the determination "YES" in the step S**404** and before the step S**406**. Therefore, in the step S**406**, the controller CU raises the edge ring ER by the supporting pins **521** in a state in which the attraction of the edge ring ER to the ring support surface **112***b* is resolved. That is, after the position misalignment amount is detected, it is possible to transfer the edge ring ER to the transfer robot TR**1** while the position of the edge ring ER is prevented from being further misaligned. An operation in the step S**407** may be the same operation as the step S**106** in FIG. **7**, and an operation in the step S**408** may be the same operation as the step S**107** in FIG. **7**.

[0165] Further, in the step S**409**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1** by the transfer robot TR**1**. The step S**409** corresponds to the step S**5** in FIG. **6**. Further, in the step S**410**, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR**1**. The step S**410** corresponds to the step S**6** in FIG. **6**.

[0166] As described above, when recognizing a misalignment of the edge ring ER in the leakage check, the transfer method according to the fourth example detects a position of the edge ring ER by the position detection sensor S**1** or S**2** of the transfer robot S**1** or S**2**. Accordingly, the controller CU may certainly detect a position misalignment amount of the edge ring ER, and thus it is possible to correct and move the edge ring ER with a high degree of precision. As a result, the transfer method may transfer the edge ring ER without interfering with other parts.

[0167] FIG. **15** is a flowchart illustrating a transfer method according to a fifth example. The transfer method according to the fifth example performs state identification and transfer of the edge ring ER by performing a leakage check in the processing module PM**1**. Further, in description of the transfer method according to the fifth example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM**1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0168] When performing the steps to the step S**3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps S**501** to S**511** as the transfer method according to the fifth example. The steps S**501** to S**511** are performed as the controller CU controls each part of the substrate processing system PS.

[0169] The steps S**501** to S**504** may be the same as the steps S**301** to S**304**. In addition, in order to perform a leakage check after the step S**502** (before the step S**503**), the controller CU performs a processing of electrostatically attracting the edge ring ER to the ring support surface **112***b*. Further, if the pressure of the pressure sensor **67** is less than the pressure threshold value in the step S**504** (the step S**504**: "NO"), the controller CU proceeds to the step S**505**. If the pressure of the pressure sensor **67** is the pressure threshold value or more, the controller CU proceeds to the step S**509** (the step S**504**: "YES").

[0170] The steps S**505** and S**506** may be the same as the steps S**205** and S**206** in the transfer method according to the second example. Further, since the leakage check is performed in a state in which the edge ring ER is electrostatically attracted, the controller CU may perform a charge removal processing of the edge ring ER before the step S**505** (or before the step S**509** to be described later).

[0171] In the step S**507**, the controller CU detects a position misalignment amount of the edge ring ER by the position detection sensors S**11** and S**12** installed in the vicinity of the gate valve (not illustrated) which partitions the vacuum transfer module TM and the processing module PM**1**. For example, when transferring the edge ring ER by the transfer robot TR**1**, the controller CU performs detection by the position detection sensors S**11** and S**12** and calculates a position of the edge ring ER based on a position at which the position detection sensors S**11** and S**12** are shaded by the edge ring ER and a time for which the position detection sensors S**11** and S**12** are shaded by the edge ring ER. Further, based on the position of the edge ring ER and a preset reference position, the controller CU calculates a position misalignment amount (an index related to the position misalignment amount) of the edge ring ER from the reference position.

[0172] After calculating the position misalignment amount when unloading the edge ring ER, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR**1** (the step S**508**). In the loading processing, when loading the edge ring ER into the accommodation module (e.g., the cassette which is not illustrated) of the ring accommodation module RSM, the controller CU sets a movement amount of the transfer robot TR**1** based on the position misalignment amount acquired in the step S**507**, and corrects and moves the transfer robot TR**1**. Accordingly, the edge ring ER is returned to the normal position by the transfer robot TR**1** without interfering with the ring accommodation module RSM. Thus, the controller CU may smoothly accommodate the edge ring ER without interfering with other parts.

[0173] Meanwhile, if a leakage amount is less than a threshold value, the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small), and hence the controller CU performs ordinary transfer (of not performing movement and correction) in the steps S**509** to S**511**. The steps S**509** to S**511** may be the same as the steps S**208** to S**210**.

[0174] As described above, when recognizing a misalignment of the edge ring ER in the leakage check, the transfer method according to the fifth example detects the position of the edge ring ER by the position detection sensors S**11** and S**12** of the vacuum transfer module TM. In this case, the controller CU may certainly detect a position misalignment amount of the edge ring ER, and thus it is possible to correct and move the edge ring ER with a high degree of precision when loading the edge ring ER into the ring accommodation module RSM.

[0175] In addition, when detecting a position of the edge ring ER by the position detection sensor S**11** and S**12** of the vacuum transfer module TM, movement and correction may be performed in the processing module PM**1** by returning the edge ring ER to the processing module PM**1** for the time being without transferring the edge ring ER to the ring accommodation module RSM after the detection.

[0176] Next, a transfer method when performing state identification and transfer of the edge ring ER by the camera CM of the transfer robot TR**1** illustrated in FIG. **5**C, is described. FIG. **16** is a flowchart illustrating a processing flow in a transfer method according to a sixth example. Further, in description of the transfer method according to the sixth example, a case where the ring **113** is transferred between the ring accommodation module RSM and the processing module PM**1** is described in detail. Even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used.

[0177] When performing the steps to the step S**3** in the operation sequence in FIG. **6**, the controller CU of the substrate processing system PS performs steps S**601** to S**609** as the transfer method according to the sixth example. The steps S**601** to S**609** are performed as the controller CU controls each part of the substrate processing system PS.

[0178] The steps S**601** and S**602** may be the same as the steps S**101** and S**102** in the transfer method according to the first example.

[0179] In the step S**603**, the controller CU acquires image information as an index related to a position misalignment amount by operating the transfer robot TR**1** and imaging the substrate support surface **112***a* and the edge ring ER by the camera CM. The controller CU performs an image processing on the image information to extract a gap amount (a position misalignment amount) between the substrate support surface **112***a* and the edge ring ER from color or contrast information.

[0180] Therefore, based on the position misalignment amount extracted from the image information and a pre-stored threshold value, the controller CU determines whether the position misalignment amount of the edge ring ER is the threshold value or more (the step S**604**). If the position misalignment amount is less than the threshold value (the step S**604**: "NO"), the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small). Therefore, the controller CU skips the steps S**605** to S**607** and immediately proceeds to the step S**608**. Meanwhile, if the position misalignment amount is the threshold value or more (the step S**604**: "YES"), the edge ring ER may be misaligned (or the position misalignment amount may be large). In this case, the controller CU detects an error of the edge ring ER, and proceeds to step S**605**.

[0181] In addition, the steps S**605** to step S**607** may be the same as the steps S**105** to S**107** in the transfer method according to the first example.

[0182] Further, in the step S**608**, the controller CU performs an unloading processing of unloading the edge ring ER from the processing module PM**1** by the transfer robot TR**1**. The step S**608** corresponds to the step S**5** in FIG. **6**. Further, in the step S**609**, the controller CU performs a loading processing of loading the edge ring ER into the ring accommodation module RSM by the transfer robot TR**1**. The step S**609** corresponds to the step S**6** in FIG. **6**.

[0183] As described above, in the transfer method according to the sixth example, it is possible to obtain the position misalignment amount of the edge ring ER by imaging the edge ring ER by the camera of the transfer robot TR**1**. Thus, the controller CU operates the transfer robot TR**1**, based on the position misalignment amount of the image information, so that it is possible to transfer the edge ring ER without interfering with other parts.

[0184] In addition, a transfer method in a configuration in which the camera CM is applied to the transfer robot TR**1** is not limited to the transfer method according to the sixth example, and may be used, for example, in combination with the leakage check or in combination with the position detection sensor S**11** and S**12** of the transfer module TM. As for the position misalignment amount of the edge ring ER, image information of the camera CM (or a detection result of the position detection sensor S**1** or S**2**) and a detection result of the position detection sensors S**11** and S**12** are used, so that it is possible to further increase precision of detection or precision of correction and movement. In addition, in the transfer method according to the sixth example, when acquiring an index related to a position misalignment amount of the edge ring ER by the position detection sensor S**1** or S**2** of the transfer robot TR**1**, detection may be performed in a state in which the edge ring ER is supported by the supporting pins **521**. That is, in the step S**601**, the controller CU may stop the supporting pins **521** at a position at which the edge ring ER is separated from the ring support surface **112***b* and acquire the position misalignment amount by the position detection sensor S**1** or S**2**, or the controller CU may acquire the position misalignment amount by the position detection sensor S**1** or S**2** at an arbitrary pin height which is higher than the position at which the edge ring ER is separated from the ring support surface **112***b* and at which the upper surface of the edge ring ER is below the position detection sensors S**1** and S**2** (see FIG. **9**).

[0185] In addition, in the configuration in which the camera CM is applied to the transfer robot TR**1**, as illustrated in FIG. **17**, an imaging direction of the camera CM is appropriately adjusted, so that it is possible not only to image the substrate support surface **112***a* or the edge ring ER below the camera CM but also to image a front in a transfer direction. Therefore, it is possible to image the edge ring ER being supported by the supporting pins **521** and obliquely inclined while getting out of some supporting pins **521** using the camera CM. Further, it is possible to image the edge ring ER is supported on the electrostatic chuck **112** while getting out of all the supporting pins **521** using the camera CM.

[0186] The controller CU image-processes and analyzes such image information to recognize that the edge ring ER is inclined or that the edge ring ER gets out of all the supporting pins. For example, the camera CM images a plurality of image information in movement of the transfer robot TR**1**, and the controller CU performs calculation by matching positions of the edge ring ER and the transfer robot TR**1** in the plurality of image information. Accordingly, the controller CU may calculate a three-dimensional form (posture and position) of the edge ring ER with a high degree of precision. Further, the controller CU may accumulate image information acquired for each image of the edge ring ER and learn recognition of the edge ring ER from images of the edge ring ER, which are included the plurality of accumulated image information. Accordingly, when acquiring image information next time, it is possible to increase extraction precision of the edge ring ER from the image information. When recognizing that the edge ring ER is inclined or that the edge ring ER gets out of all the supporting pins, the controller CU may be configured to issue an alarm.

[0187] FIG. **18** is a view illustrating a modification of FIG. **5**B, which is a configuration of performing a leakage check of gas as an index related to a position misalignment amount of the edge ring. As illustrated in FIG. **18**, the configuration of performing the leakage check may be a configuration including a groove **61***ag* that communicates with the gas supply port **61***a* while being recessed downward in the ring support surface **112***b* of the peripheral edge of the electrostatic chuck **112**. The groove **61***ag* is formed in an annular shape that goes around in a circumferential direction of the ring support surface **112***b*. According to the configuration described above, the groove **61***ag* may supply a gas such as He gas to the entire rear surface of the inner ring **113***a* in the circumferential direction to perform a leakage check of the gas.

[0188] FIG. **19** is a flowchart illustrating an operation sequence of transfer of the edge ring ER according to a modification. In the modification shown in FIG. **19**, the substrate processing system PS and the transfer method may detect a misalignment amount of the edge ring ER by the position detection sensors S**11** and S**12** of the vacuum transfer module TM when unloading the edge ring ER from the processing module PM**1**. Further, the substrate processing system PS loads the edge ring ER into the ring accommodation module RSM while correcting a position of the edge ring ER based on the misalignment amount of the edge ring ER, which is detected by the position detection sensors S**11** and S**12**. In addition, even in a case where a processing module for an exchange target is any one of the processing modules PM**2** to PM**7**, the same method as a case where the processing module for the exchange target is the processing module PM**1** may be used. In this case, a position detection sensor located at a position adjacent to each processing module may be used.

[0189] The detection of the misalignment amount of the edge ring ER by the position detection sensors S**11** and S**12** may be executed by a method shown in FIGS. **20**A and **20**B. FIG. **20**A is a view illustrating a relationship between a position of the edge ring ER and positions of the position detection sensors S**11** and S**12**. FIG. **20**B is a view illustrating a change in sensor output of the position detection sensors S**11** and S**12** when transferring the edge ring ER from a position P**21** to a position P**24**. Further, in FIG. **20**B, a time at the position P**21** represents t**21**, a time at the position P**22** represents t**22**, a time at the position P**23** represents t**23**, and a time at the position P**24** represents t**24**. Based on a position of the edge ring ER, which is detected by the position detection sensors S**11** and S**12**, and a preset reference position, the controller CU calculates a misalignment amount of the edge ring ER from the reference position. Then, the controller CU loads the edge ring ER into the ring accommodation module RSM by the transfer robot TR**1** to correct the calculated misalignment amount. Accordingly, even if a position of the edge ring ER held by the upper fork FK**1** or the lower fork FK**2** is misaligned from the reference position, it is possible to load the edge ring ER at a predetermined position of the ring accommodation module RSM.

[0190] The position of the edge ring ER held by the upper fork FK**1** or the lower fork FK**2** may be calculated based on an output change of the position detection sensors S**11** and

S12, which is generated as an inner peripheral edge passes through the position detection sensors S11 and S12. For example, as illustrated in FIG. 20A, when transferring the edge ring ER from the position P24 to the position P24, the position of the edge ring ER may be calculated based on a time T2 for which the edge ring ER moves from the position P22 to the position P23. The position P22 is a position at which the sensor output of the position detection sensor S11 and S12 is changed from a low (L) level to a high (H) level, and the position P23 is a position at which the sensor output of the position detection sensors S11 and S12 is changed from the high (H) level to the low (L) level. Specifically, as illustrated in FIG. 20B, the position of the edge ring ER may be calculated by T2=t23−t22 using the time t22 at the position P22 and the time t23 at the position P23. Further, in FIGS. 20A and 20B, a case where a position at which the position detection sensor S11 is shaded by the edge ring ER and a position at which the position detection sensor S2 is shaded by the edge ring ER are the same is shown, but the positions may be different from each other.

[0191] The controller CU controls steps S701 to S706 in FIG. 19. First, the controller CU raises the supporting pins 521 of the plasma processing apparatus 1, and allows the edge ring ER to be spaced apart from the ring support surface 112*b* of the substrate support 11 (the step S701 in FIG. 19). When residual attraction exists between the ring support surface 112*b* and the edge ring ER, a misalignment may occur in the edge ring ER when raised.

[0192] Subsequently, the controller CU enters the transfer robot TR1 below the edge ring ER in the plasma processing space 10*s*, and transfers the edge ring ER to the transfer robot TR1 by lowering the supporting pin 521 (the step S702 in FIG. 19).

[0193] After that, the controller CU unloads the transfer robot TR1 from the processing module PM1, and detects a position of the edge ring ER held by the transfer robot TR1 by the position detection sensors S11 and S12 (the step S703). The transfer robot TR1 moves to pass through a middle of the position detection sensors S11 and S12, and the center of the edge ring ER passes the middle when the edge ring ER is not misaligned. Meanwhile, when the edge ring ER is misaligned, a misalignment of the edge ring ER (a position misalignment amount and a position misalignment direction of the center of the edge ring ER with respect to a reference position of the transfer robot TR) may be calculated by the position detection sensors S11 and S12.

[0194] Subsequently, the controller CU compares the position misalignment amount of the edge ring ER with a pre-stored threshold value, and determines whether the position misalignment amount of the edge ring ER is the threshold value or more (the step S704). If the position misalignment amount is less than the threshold value (the step S704: "NO"), the edge ring ER may not be misaligned (or the position misalignment amount may be sufficiently small). In this case, the controller CU proceeds to the step S705.

[0195] In the step S705, the controller CU loads the edge ring ER into the ring accommodation module RSM by transferring the edge ring ER without correcting movement of the transfer robot TR1. Accordingly, the substrate processing system PS may smoothly load the edge ring ER into the ring accommodation module RSM.

[0196] Meanwhile, if the position misalignment amount is the threshold value or more (the step S704: "YES"), the edge ring ER may be misaligned (or the position misalignment amount may be large). In this case, the controller CU proceeds to the step S706 in FIG. 19 to load the edge ring ER into the ring accommodation module RSM while correcting the movement of the transfer robot TR1 based on the misalignment of the edge ring ER, which is calculated from a detection result of the position detection sensors S11 and S12. At this time, the controller CU sets a movement amount and a movement direction in the correction and movement of the transfer robot TR1 based on the acquired misalignment amount and the acquired misalignment direction, and corrects the movement of the transfer robot TR1. Accordingly, even if a misalignment occurs in the edge ring ER, the substrate processing system PS may smoothly load the edge ring ER into the ring accommodation module RSM.

[0197] In addition, the transfer method in FIG. 19 may be applied to even a case where the cover ring CR is unloaded from any one of the processing modules PM1 to PM7 to the vacuum transfer module TM and transferred into the ring accommodation module RSM. Further, the transfer method in FIG. 19 may be applied to even a case where the upper ring 222 or the lower ring 221 in FIG. 11 is unloaded from any one of the processing modules PM1 to PM7 to the vacuum transfer module TM and transferred to the load lock module LLM. Further, the transfer method in FIG. 19 may be applied to even a case where the upper ring 222 or the lower ring 221 in FIG. 11 is unloaded from any one of the processing modules PM1 to PM7 to the vacuum transfer module TM and transferred to the ring accommodation module RSM.

[0198] Alternatively, as another modification, the substrate processing system PS and the transfer method may detect a position of the edge ring by the position detection sensor S1 or S2, by raising the edge ring ER and then lowering the edge ring ER by the supporting pins 521, to place the edge ring ER on the ring support surface 112*b*. If a position misalignment amount of the edge ring ER is smaller than a threshold value, the substrate processing system PS and the transfer method may perform an unloading processing from the processing module PM1 (unloading while detecting a position of the edge ring ER by the position detection sensors S11 and S12, or the like) by raising the edge ring ER again by the supporting pins 521 and transferring the edge ring ER to the transfer robot TR1, and perform a loading processing into the ring accommodation module RSM (correction and movement based on the position misalignment amount). In the unloading processing of the edge ring ER from the processing module PM1 and the loading processing of the edge ring ER into the ring accommodation module RSM, the processing flow shown in FIG. 19 may be performed. Meanwhile, when the position misalignment amount of the edge ring ER is the threshold value or more, the substrate processing system PS and the transfer method may perform a processing of taking out the edge ring ER by issuing an alarm and opening the plasma processing chamber 10 in an atmosphere.

[0199] The embodiment disclosed in the above includes, for example, the following aspect.

(Supplementary Note 1)

[0200] A substrate processing system including:

[0201] a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring;

[0202] a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and

[0203] a controller,

[0204] wherein the controller performs:

[0205] (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support;

[0206] (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and

[0207] (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

(Supplementary Note 2)

[0208] The substrate processing system of Supplementary Note 1, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor included in the transfer robot.

(Supplementary Note 3)

[0209] The substrate processing system of Supplementary Note 2, wherein, in step (B), the controller acquires a gap amount of the ring in a horizontal direction with respect to a substrate support surface of the substrate support based on a position of the substrate support surface and the position of the ring, which are detected by the position detection sensor.

(Supplementary Note 4)

[0210] The substrate processing system of any one of Supplementary Notes 1 to 3, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, a leakage amount of a gas by supplying the gas between the support surface of the substrate support and a rear surface of the ring in a state in which the ring is electrostatically attracted to the support surface.

(Supplementary Note 5)

[0211] The substrate processing system of any one of Supplementary Notes 1 to 4, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, image information of the ring, which is imaged by a camera included in the transfer robot.

(Supplementary Note 6)

[0212] The substrate processing system of any one of Supplementary Notes 1 to 5, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor installed in the vacuum transfer module, when unloading the ring from the processing module by the transfer robot.

(Supplementary Note 7)

[0213] The substrate processing system of any one of Supplementary Notes 1 to 6, wherein, if it is determined to perform the correction in step (C), the controller further performs correction and movement of moving the ring according to the position misalignment amount by the transfer robot, and

[0214] wherein, if it is determined not to perform the correction in step (C), the controller further performs: transferring the ring by the transfer robot, without performing the correction and movement.

(Supplementary Note 8)

[0215] The substrate processing system of Supplementary Note 7, wherein, if it is determined to perform the correction in step (C), the controller performs the correction and movement by the transfer robot in the processing module.

(Supplementary Note 9)

[0216] The substrate processing system of Supplementary Note 8, wherein, in the correction and movement, the controller sequentially performs:

[0217] receiving the ring in which a misalignment occurs from the lifter by the transfer robot;

[0218] moving the transfer robot according to the position misalignment amount; and

[0219] transferring the ring from the transfer robot to the lifter.

(Supplementary Note 10)

[0220] The substrate processing system of Supplementary Note 8, wherein, in the correction and movement, the controller performs: moving the transfer robot according to the position misalignment amount before receiving the ring from the lifter by the transfer robot.

(Supplementary Note 11)

[0221] The substrate processing system of Supplementary Note 7, wherein, if it is determined to perform the correction in step (C), the controller further performs: transferring the ring to a ring accommodation module configured to accommodate the ring while performing the correction and movement on the ring in which a misalignment occurs by the transfer robot.

(Supplementary Note 12)

[0222] The substrate processing system of any one of Supplementary Notes 1 to 3 and 5, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is placed on the substrate support by lowering the lifter.

(Supplementary Note 13)

[0223] The substrate processing system of any one of Supplementary Notes 1 to 3 and 5, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is raised by the lifter in step (A).

(Supplementary Note 14)

[0224] The substrate processing system of any one of Supplementary Notes 1 to 13, further including: a first ring accommodation module connected to the vacuum transfer module and configured to accommodate the ring,

[0225] wherein the controller further performs: loading the ring unloaded from the processing module into the first ring accommodation module by the transfer robot.

(Supplementary Note 15)

[0226] The substrate processing system of any one of Supplementary Notes 1 to 13, further including:

[0227] an atmospheric transfer module connected to the vacuum transfer module via a load lock module; and

[0228] a second ring accommodation module connected to the atmospheric transfer module and configured to accommodate the ring,

[0229] wherein the controller further performs: loading the ring unloaded from the processing module into the second ring accommodation module via the load lock module and the atmospheric transfer module by the transfer robot.

(Supplementary Note 16)

[0230] The substrate processing system of any one of Supplementary Notes 1 to 15, wherein the processing module performs a substrate processing in a state in which the ring is electrostatically attracted, and

[0231] wherein, before step (A), the controller further performs: removing charges from the ring.

(Supplementary Note 17)

[0232] The substrate processing system of any one of Supplementary Notes 1 to 16, wherein the controller further performs: cleaning of an interior of the processing chamber by generating plasma during step (A) or before and after step (A).

(Supplementary Note 18)

[0233] The substrate processing system of Supplementary Note 17, wherein the cleaning of the interior of the processing chamber during step (A) or after step (A) is performed in a state in which the ring is spaced apart from the support surface of the substrate support.

(Supplementary Note 19)

[0234] The substrate processing system of any one of Supplementary Notes 1 to 18, wherein the lifter includes a supporting pin and an actuator configured to vertically move the supporting pin.

(Supplementary Note 20)

[0235] A transfer method of unloading a ring disposed at a periphery of a substrate from a processing module by a transfer robot of a vacuum transfer module connected to the processing module, wherein the processing module includes a processing chamber, a substrate support configured to support the substrate and the ring in the processing chamber, and a lifter configured to raise and lower the ring, the transfer method including:

[0236] (A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support;

[0237] (B) after step (A), acquiring an index related to a position misalignment amount of the ring; and

[0238] (C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

[0239] In addition, the present disclosure is not limited to the configurations shown herein, including configurations exemplified in the embodiments, combinations with other elements, and the like. In this regard, this may be changed without departing from the spirit of the present disclosure, and may be properly set according to an application form thereof. Further, in items described in a plurality of embodiments, other configurations may be used within a non-contradictory range and may be combined within a non-contradictory range.

[0240] For example, in the embodiment, a capacitively coupled plasma apparatus is described as an example. However, the present disclosure is not limited thereto, and may be applied to other plasma apparatuses. For example, instead of the capacitively coupled plasma apparatus, an inductively coupled plasma (ICP) apparatus may be used. In this case, the inductively coupled plasma apparatus includes an antenna and a lower electrode. The lower electrode is disposed in a substrate support, and the antenna is disposed on or above a chamber. Further, an RF generator is coupled to the antenna, and a DC generator is coupled to the lower electrode. Therefore, the RF generator is coupled to an upper electrode of the capacitively coupled plasma apparatus or the antenna of the inductively coupled plasma apparatus. That is, the RF generator is coupled to the plasma processing chamber **10**.

[0241] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-174836, filed on Oct. 31, 2022, the entire contents of which are incorporated herein by reference.

[0242] According to the present disclosure in some embodiments, it is possible to suppress interference between a ring and other parts when transferring the ring.

[0243] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

What is claimed is:

1. A substrate processing system comprising:

a processing module including a processing chamber, a substrate support configured to support a substrate and a ring disposed at a periphery of the substrate in the processing chamber, and a lifter configured to raise and lower the ring;

a vacuum transfer module connected to the processing module and including a transfer robot configured to transfer the ring; and

a controller,

wherein the controller performs:

(A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support;

(B) after step (A), acquiring an index related to a position misalignment amount of the ring; and

(C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

2. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor included in the transfer robot.

3. The substrate processing system of claim **2**, wherein, in step (B), the controller acquires a gap amount of the ring in a horizontal direction with respect to a substrate support surface of the substrate support based on a position of the substrate support surface and the position of the ring, which are detected by the position detection sensor.

4. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, a leakage amount of a gas by supplying the gas between the support surface of the substrate support and a rear surface of the ring in a state in which the ring is electrostatically attracted to the support surface.

5. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, image information of the ring, which is imaged by a camera included in the transfer robot.

6. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires, as the index related to the position misalignment amount, the position of the ring, which is detected by a position detection sensor installed in the vacuum transfer module, when unloading the ring from the processing module by the transfer robot.

7. The substrate processing system of claim **1**, wherein, if it is determined to perform the correction in step (C), the controller further performs correction and movement of moving the ring according to the position misalignment amount by the transfer robot, and

wherein, if it is determined not to perform the correction in step (C), the controller further performs: transferring the ring by the transfer robot, without performing the correction and movement.

8. The substrate processing system of claim **7**, wherein, if it is determined to perform the correction in step (C), the controller performs the correction and movement by the transfer robot in the processing module.

9. The substrate processing system of claim **8**, wherein, in the correction and movement, the controller sequentially performs:

receiving the ring in which a misalignment occurs from the lifter by the transfer robot;

moving the transfer robot according to the position misalignment amount; and

transferring the ring from the transfer robot to the lifter.

10. The substrate processing system of claim **8**, wherein, in the correction and movement, the controller performs: moving the transfer robot according to the position misalignment amount before receiving the ring from the lifter by the transfer robot.

11. The substrate processing system of claim **7**, wherein, if it is determined to perform the correction in step (C), the controller further performs: transferring the ring to a ring accommodation module configured to accommodate the ring while performing the correction and movement on the ring in which a misalignment occurs by the transfer robot.

12. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is placed on the substrate support by lowering the lifter.

13. The substrate processing system of claim **1**, wherein, in step (B), the controller acquires the index related to the position misalignment amount in a state in which the ring is raised by the lifter in step (A).

14. The substrate processing system of claim **1**, further comprising: a first ring accommodation module connected to the vacuum transfer module and configured to accommodate the ring,

wherein the controller further performs: loading the ring unloaded from the processing module into the first ring accommodation module by the transfer robot.

15. The substrate processing system of claim **1**, further comprising:

an atmospheric transfer module connected to the vacuum transfer module via a load lock module; and

a second ring accommodation module connected to the atmospheric transfer module and configured to accommodate the ring,

wherein the controller further performs: loading the ring unloaded from the processing module into the second ring accommodation module via the load lock module and the atmospheric transfer module by the transfer robot.

16. The substrate processing system of claim **1**, wherein the processing module performs a substrate processing in a state in which the ring is electrostatically attracted, and

wherein, before step (A), the controller further performs: removing charges from the ring.

17. The substrate processing system of claim **1**, wherein the controller further performs: cleaning of an interior of the processing chamber by generating plasma during step (A) or before and after step (A).

18. The substrate processing system of claim **17**, wherein the cleaning of the interior of the processing chamber during step (A) or after step (A) is performed in a state in which the ring is spaced apart from the support surface of the substrate support.

19. The substrate processing system of claim **1**, wherein the lifter includes a supporting pin and an actuator configured to vertically move the supporting pin.

20. A transfer method of unloading a ring disposed at a periphery of a substrate from a processing module by a transfer robot of a vacuum transfer module connected to the processing module, wherein the processing module includes a processing chamber, a substrate support configured to support the substrate and the ring in the processing chamber, and a lifter configured to raise and lower the ring, the transfer method comprising:

(A) raising the lifter to allow the ring to be spaced apart from a support surface of the substrate support;

(B) after step (A), acquiring an index related to a position misalignment amount of the ring; and

(C) determining whether to correct a position of the ring based on the index related to the position misalignment amount, which is acquired in step (B).

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